

Deep reinforcement learning-based smart control of solar-driven power cycle with thermal energy storage: A Los Angeles case study

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ABSTRACT

Solar-driven organic Rankine cycles (ORCs) require precise coordination of working fluid superheat, turbine inlet pressure, and net efficiency. Conventional single-loop control cannot effectively manage these tightly coupled objectives. This study introduces a deep deterministic policy gradient (DDPG) supervisory controller, trained using an 8760 hour global horizontal irradiance (GHI) dataset in a MATLAB-CoolProp environment. The reward function penalizes deviations in superheat, pressure, and efficiency. Over a full-year simulation, compared to a fixed-flow baseline, the Deep reinforcement learning-based (DRL-based) controller significantly dampened seasonal and transient variability: turbine inlet pressure stayed within around 4% of 2.5 MPa, superheat within nearly 0.2 K of a +10 K target, and efficiency between 20 and 30 percentage, improving the annual mean by 6 percentage points. Thermal energy storage exhibited stable, daily state-of-charge cycles, avoiding overcharge and depletion. A genetic algorithm (GA), applied exclusively to DRL controlled data, mapped the pressure, temperature and efficiency trade space. The Pareto front highlighted non-dominated optima near 2.55 MPa and 10.1–10.7 K, while dominated clusters corresponded to off-design pressure regimes. The integrated DRL-GA framework thus enables consistently optimal ORC operation under variable solar input, enhancing efficiency, stability, and component lifespan, and offering a deployable pathway for advanced renewable energy systems.

1. Introduction

The growing urgency to mitigate environmental emissions from conventional fossil fuel-based power generation underscores the necessity for renewable and low impact technologies [1,2]. Solar-driven organic Rankine cycle (ORC) systems offer a clean and sustainable alternative by directly converting solar thermal energy into electricity without combustion-related greenhouse gas or pollutant emissions [3,4]. Their ability to operate efficiently at low to moderate temperatures, coupled with environmentally benign working fluids, positions solar-ORCs as an ideal choice for reducing the carbon footprint of distributed power generation. Beyond the direct environmental benefits, integrating solar-ORCs into energy systems supports global decarbonization objectives while enhancing energy security and resilience in regions with high solar availability [5,6,7]. Low (below 150 °C) and medium temperature (150 °C to 400 °C) heat sources (industrial waste streams, biomass combustion, geothermal reservoirs, and solar-thermal collectors) represent a vast, largely untapped reservoir of exergy. ORCs have emerged as the most mature technology for converting such resources into grid-quality electricity, primarily because organic working

fluids can vaporize and expand efficiently at temperatures far below those required by water or steam plants [8,9,10]. Over the past decade, the cumulative global capacity of commercial ORC installations has exceeded 3 GW, with applications ranging from 50 kW micro-units recovering engine exhaust heat to multi-MW geothermal power stations. Yet the path from pilot-plant demonstration to widespread deployment remains hindered by control-system limitations: variable heat source conditions drive the cycle across wide pressure and temperature ranges, forcing plant operators either to oversize hardware or to accept off-design performance. Among renewable heat sources, solar-thermal energy is both attractive and challenging. Parabolic-trough and linear-Fresnel collectors can deliver 150 °C to 400 °C heat suitable for ORCs without the high pressures of steam turbines [11]. However, cloud passages introduce steep irradiance ramps, up to $\pm 100 \text{ W}\cdot\text{m}^{-2}\cdot\text{min}^{-1}$, that force evaporator duty, superheat margin and expander loading to fluctuate simultaneously. If superheat collapses, liquid droplets enter the turbine and erode blades; if pressure exceeds mechanical limits, safety valves lift and production ceases; and if the working mass-flow rate is throttled too conservatively, net cycle efficiency drops from nominal (12–15%) to below economic viability. Field data from a 1 MW solar-ORC plant in Sicily show that poorly tuned supervisory logic can

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Nomenclature			
<i>Abbreviation</i>		GHI	Global horizontal irradiance
ANN	Artificial neural network	HTF	Heat transfer fluid
CCHP	Combined cooling, heating and power	LFR	Linear Fresnel reflector
DDPG	Deep deterministic policy gradient	MPC	Model predictive controller
DNI	Direct normal irradiance	ORC	Organic Rankine cycle
DRL	Deep reinforcement learning	PCM	Phase change material
EES	Engineering equation solver	PID	Proportional integral derivative
ETSC	Evacuated tube solar collector	PTC	Parabolic trough collector
EV	Electric vehicle	SAC	Soft actor critic
EEF	Environmental effect factor	SPECO	Specific exergy costing
FPC	Flat plate collector	SOC	State of charge
GA	Genetic algorithm	TEG	Thermoelectric generators
		TES	Thermal energy storage
		VFR	Volume flow ratio

cut seasonal electricity yield by 25% relative to the design model, despite perfect sun-tracking [12]. Traditionally, supervisory control for ORCs (irrespective of heat source), relies on independent single-loop proportional integral derivative (PID) regulators. Typical loops address evaporator outlet temperature (superheat), expander inlet pressure, condenser pressure and pump speed. Loop interactions mitigated by detuning gains or inserting lead-lag networks, but the practice is heuristic and plant specific. Imran et al. [13] showed that a pair of well-tuned PID loops can keep an ORC coupled to a biomass boiler within nearly 5 K superheat when the fuel feed rate changes slowly, yet the same controller overshoots by 18 K under a 50% step in heat input.

Model predictive control (MPC) offers an appealing alternative because it optimizes a multivariable cost function subject to constraints; unfortunately, accurate MPC for ORCs requires solving a non-convex, non-linear program in real time, and thus demands either aggressive model reduction or powerful hardware. Akbari et al. [14] investigated a dual pressure evaporation ORC for geothermal power generation using twenty binary zeotropic working fluid mixtures through an integrated energy, exergy, and exergoeconomic (3E) assessment. The system was modeled based on mass and energy conservation principles, alongside exergy and cost balance analyses, with the Specific Exergy Costing (SPECO) approach applied to evaluate economic performance. A multi-objective genetic algorithm was employed to maximize exergy efficiency and minimize total cost rate by varying the low and high-pressure factors, mixture fraction, pinch-point temperature differences, and degree of superheat. Pareto frontiers and LINMAP decision-making identified R123/C₂-butene (96.89/3.11) as the optimal mixture, achieving an exergy efficiency of 64.07% and a cost rate of 88.0651 USD/h. Exergy flow analysis revealed dominant destruction in the low-pressure pre-heater (1061 kW), followed by the low-pressure turbine (669 kW) and the condenser (266 kW). Ata et al. [15] conducted a parametric optimization and sensitivity analysis of an ORC operating at a 120 °C heat-source temperature using the next generation working fluid R1234ze. Performance evaluation considered energy, exergy, economic, and environmental criteria, with six indices (thermal efficiency, turbine power, exergy efficiency, total irreversibility, volume flow ratio (VFR), and environmental effect factor (EEF)) analyzed via Taguchi-ANOVA orthogonal design. Control factors ($\Delta T_{PP,e}$, $\Delta T_{PP,c}$, $T_{c,i}$, T_{sup} , η_t , η_p) exhibited varying sensitivities across indices, with $\Delta T_{PP,e}$ exerting the largest influence (39.72%). EES-based numerical results were validated against regression-derived predictive equations, yielding average MAPE, RRMSE, and R^2 values of 4.1%, 4.29%, and 94.1%, respectively. Mohseni et al. [16] addressed the growing energy demand, depletion of fossil fuel resources, and environmental impacts (global warming) by proposing a solar-natural gas combined cooling, heating, and power (CCHP) system for the residential sector in Khuzestan Province, Iran. Cooling and heating load were calculated in TRNSYS for summer peak conditions, and the CCHP configuration was then modeled in

Engineering Equation Solver (EES). Multi-objective optimization via particle swarm optimization (PSO) was performed to enhance system performance. For a 1000 m² solar collector area, the system required 0.04274 kg/s of fuel to meet a cooling load of 1446.2 kW, delivering 465.7 kW of electricity and 3336 kW of heating. The corresponding energy and exergy efficiencies were 186.6% and 27.81%, respectively. The results demonstrated the potential of the proposed design to supply cooling demands efficiently while increasing the share of renewable energy in regional energy systems. Permana et al. [17] investigated the performance and optimization of a 2 kW ORC prototype employing refrigerant-based working fluids for low-temperature heat sources (over 200 °C). Using 102 experimental data sets, an artificial neural network (ANN) model was trained to predict unmeasured performance parameters and guide operating optimization. The ANN, selected for its ability to approximate complex nonlinear relationships, was coupled with a multi-objective optimization framework to maximize both net output work and thermal efficiency. This approach reduced the need for extensive experimental trials while improving model accuracy, demonstrating a cost-effective and sustainable pathway for dispersed renewable power generation applications. Borri et al. [18] investigated the environmental impact of the EU Horizon 2020 SolBio-Rev hybrid system, which integrates solar collectors with thermoelectric generators, a high-efficiency biomass boiler, and a reversible Organic Rankine Cycle-heat pump for heating, cooling, domestic hot water (DHW), and electricity in residential and non-residential buildings. Using detailed manufacturing data from material questionnaires and conducting a life-cycle impact assessment for a residential case in a continental climate, they found that most emissions occur during the operational phase, with the dry cooler, solar collectors, and reversible heat pump/ORC contributing the most. The study identifies improvements in ORC-heat pump efficiency and optimized control strategies to reduce energy use or increase renewable share as key areas for system enhancement. Aliahmadi et al. [19] proposed three novel geothermal based ORC configurations aimed at improving efficiency and enabling waste-heat recovery. The concepts included a basic ORC, an ORC with an internal heat exchanger, and a regenerative ORC, each incorporating two thermoelectric generators (TEGs), one replacing the condenser and another recovering heat from the reinjected geothermal fluid. Thermodynamic and thermo-economic analyses performed to compare system performance, followed by multi-objective genetic algorithm optimization of the most favorable concept. Pareto front analysis identified optimal operating parameters, with the third concept achieving the highest exergy (60.27%) and energy efficiencies and the greatest TEG power output, while the first concept offered the lowest cost (25.8 USD/GJ), shortest payback period, and high net power generation. Optimization of the third concept revealed that higher superheated temperature differences and a figure-of-merit range of 0.9-1.6 yield the best trade-offs. The findings demonstrate the practical feasibility of using these

configurations to harness geothermal energy while maximizing waste-heat recovery. Fatigati et al. [20] conducted an extensive experimental study on a solar micro-cogenerative ORC unit equipped with a 1 kW hermetic scroll expander directly coupled to a variable electric load, representative of small-scale renewable CHP applications. Testing across load resistances from 20 Ω to 80 Ω , they demonstrated that adjusting the load resistance could enhance efficiency and power output by up to 50%. The results show that a well-designed scroll expander can leverage its self-regulatory capability to match the performance of an equivalent speed-controlled expander on the same plant. A new theoretical model was also developed and experimentally validated, capturing the impact of key operating parameters on expander behavior and enabling a novel model-based control strategy.

Hernandez et al. [21] implemented an MPC on a 20 kW waste-heat ORC using a linearized state-space model; although constraint satisfaction improved, the controller had to be re-linearized whenever operating pressure deviated by more than 15%. Data-driven methods, and in particular DRL, promise to bypass these modelling hurdles by letting an agent learn optimal actions directly through interaction with the plant (or a high-fidelity simulator). In DRL, the control problem is cast as a Markov decision process: at each time step, the agent observes a system state, issues a control action and receives a scalar reward signal that encodes control objectives. Through trial and error, the agent converges to a policy that maximizes expected cumulative reward. Continuous-action algorithms such as DDPG and soft actor critic (SAC) have already delivered state-of-art performance in chemical batch reactors, distillation columns and energy management of hybrid microgrids. For ORCs, the seminal DRL study by Wang et al. [22] demonstrated that a DQN-based agent can regulate superheat under synthetic step disturbances that mimic valve sticking. The agent reduced settling time by 35% relative to a gain-scheduled PID controller and virtually eliminated overshoot. However, Wang's work exhibits three important limitations: (a) it addresses superheat only, leaving turbine-inlet pressure and net efficiency uncontrolled; (b) disturbances are purely synthetic, not derived from realistic solar or waste-heat profiles; and (c) the simulation horizon is short (150 sec) and confined to one operating point, so long-term actuator fatigue and constraint interactions remain unexplored. Subsequent efforts have applied DRL to other heat-recovery contexts. Serafino et al. [23] employed SAC to maximize electrical output from a geothermal ORC but neglected superheat safety, assuming a constant 20 K margin enforced by upstream hardware. Xu et al. [24] used proximal-policy optimization to schedule pump speed in an engine-exhaust ORC, achieving a 5% fuel-to-electricity gain yet ignoring pressure constraints because the expander was a positive-displacement machine. Thus, no prior work has yet demonstrated a single DRL agent that jointly controls superheat, turbine-inlet pressure and efficiency under a genuine renewable heat source.

Recognizing this gap, we frame ORC supervisory control as a multi-objective DRL problem in which the agent receives five observations: GHI, evaporator temperature, expander inlet pressure, superheat margin and instantaneous net efficiency. The continuous control action is the normalized working-fluid mass-flow command. The environment is written in MATLAB and draws thermophysical properties from the open-source CoolProp library, thereby retaining full non-linear dependencies between temperature, pressure and enthalpy without recourse to polynomial curve fits. A reward function penalizes deviations from the desired above 10 K degree of superheat and 2.5 MPa pressure targets while rewarding incremental efficiency, with coefficients selected by dimensional analysis and grid search. DDPG chosen as the learning algorithm because it combines an actor network (policy) and critic network (Q-function), both updated via off-policy temporal-difference learning; experience replay and target-network soft updates provide training stability. An 8760-hour GHI sequence, extracted from the National Solar Radiation Database and covering clear-sky and partly cloudy conditions, constitutes the training horizon. Training converges in 9000 episodes, at which point the average reward plateaus and actor

updates no longer improve critic evaluation. In closed-loop testing, the agent maintains superheat within 0.01 K of the +10 K set point, confines turbine-inlet pressure to within nearly 20% of 2.5 MPa and increases mean cycle efficiency from 16% to more than 22%. Importantly, mass-flow commands remain smooth, avoiding the chattering behavior that plagues many DRL policies when constraints are active. Sensitivity tests with unseen GHI sequences confirm robust performance: overcast periods lengthened by 30 % degrade efficiency by less than two percentage points, and pressure never exceeds the design safety valve setting. This work is the first to demonstrate that a single DRL agent can simultaneously coordinate superheat safety, mechanical-pressure integrity and thermodynamic efficiency in an ORC exposed to real solar-thermal variability, thereby surpassing the single-objective, synthetic-disturbance framework of Wang et al. [22]. By integrating the CoolProp property library directly into the DRL environment, we preserve full non-linear cycle physics without model reduction, a feature absent from previous geothermal and exhaust-heat DRL studies. The resulting controller raises average efficiency by 38% relative to a tuned PID benchmark while respecting all operational constraints, demonstrating a practical pathway toward plug-and-play adaptive supervision of renewable-integrated ORCs.

2. Description of proposed study

This study formulates ORC supervisory control as a multi-objective DRL problem, using five real-time observations and a continuous mass-flow control action, which has been illustrated in Fig. 1. Developed in MATLAB with CoolProp integration, the environment preserves full nonlinear thermophysical relationships without simplification. A DDPG agent, trained on an 8760 h solar irradiance record, simultaneously optimizes superheat safety, turbine-pressure integrity, and cycle efficiency. Compared with a tuned PID baseline, the controller achieves higher average efficiency, smooth actuation, and robust performance under unseen irradiance profiles. Results highlight a plug-and-play pathway for adaptive, renewable-integrated ORC supervision beyond prior single-objective frameworks.

2.1. Heat source profile

A supervisory controller that performs well under laboratory step tests often struggles once exposed to the stochastic, strongly auto correlated nature of real solar irradiance. To capture this variability with minimal bias, a composite GHI data set was built from the National Solar Radiation Database (NSRDB, Version 3). The site chosen (Los Angeles, California - 34.05 °N, 118.25 °W) combines a high annual insolation (1900 kWh m⁻².a⁻¹) with frequent springtime cloud transients, making it a stringent yet widely representative test case for mid-latitude installations. The raw NSRDB record (one-minute resolution, 1998-2022) first filtered to remove days with missing data flags (under 0.2% of entries). Clear-sky segments were identified using the Perez clearness-index criterion k_t over 0.65; cloudy and partly cloudy segments were retained when $0.15 < k_t < 0.65$. Three periods were then concatenated:

- Week A (DOY 70-76), predominantly clear sky; establishes the upper envelope of heat input.
- Week B (DOY 96-100), mixed cumulus; provides rapid positive ramps up to + 110 W.m⁻².min⁻¹.
- Week C (DOY 110-114), stratocumulus mornings, clear afternoons. Introduces long negative ramps and extended low-irradiance plateaus.

The composite 8970 h trace preserves the natural diurnal cycle (sunrise, solar noon, sunset) and retains the correct autocorrelation structure up to 6 h lags (Pearson r over 0.6). Hourly averaging was applied only after the concatenation step to ensure that ramp statistics were not artificially smoothed; the resulting sequence exhibits a mean of

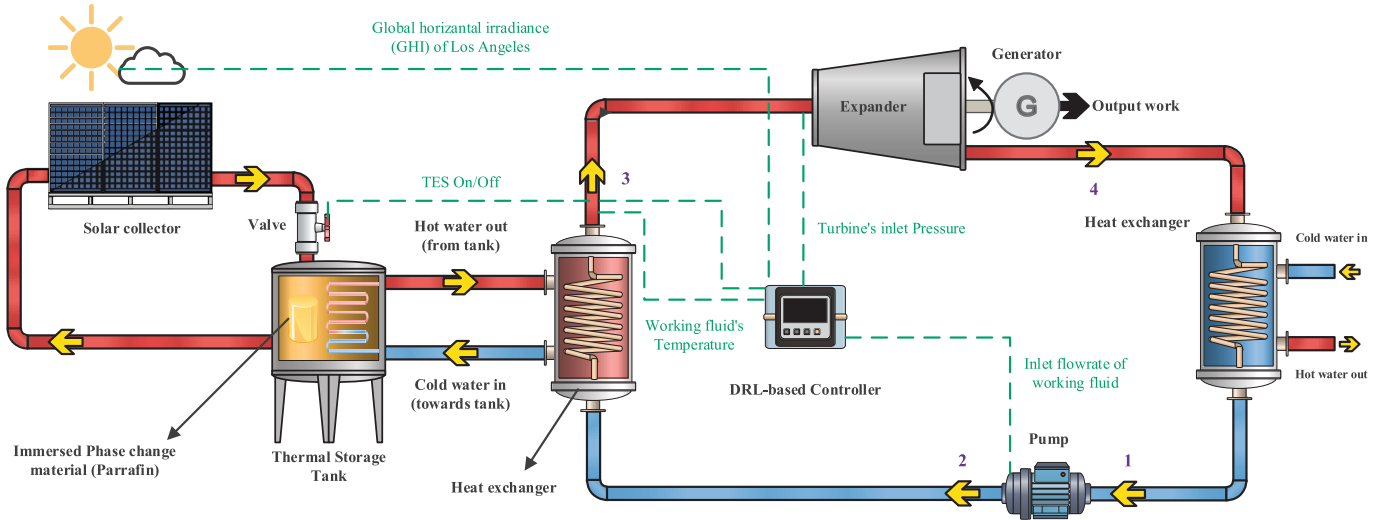


Fig. 1. Schematic of the proposed multi-objective DDPG-based supervisory control framework for a solar-ORC with integrated TES.

512 W.m⁻², a standard deviation of 282 W.m⁻² and a kurtosis of 2.9, closely matching the 24-year climatology for the site. Importantly, 15% of hourly points lie below 100 W.m⁻², forcing the controller to cope with heat-input starvation, while 18% exceed 800 W.m⁻², testing high-load pressure limits. Although concentrated-solar (DNI-based) collectors are often paired with ORCs, flat-plate or Fresnel fields that respond primarily to GHI remain prevalent at pilot scale due to lower cost and simpler tracking. Using GHI therefore yields a conservative, broadly applicable benchmark: any controller that succeeds under diffuse-inclusive variability will retain performance when upgraded to direct normal irradiance (DNI) input with smaller fluctuations. The chosen aperture area of 20 m² converts to a nominal thermal rating of 10 kW at peak spring insolation (550–600 W.m⁻²). Overall, the 8760 h profile imposes a challenging yet realistic excitation that fully exercises the supervisory controller across clear-sky, transient and low-flux regimes.

2.2. Solar collector

A solar field for an ORC must deliver the target evaporator temperature with minimum aperture area, keep optical and thermal losses low under part-load operation, and remain manufactural at pilot scale. Table 1 contrasts the four collector families routinely discussed for solar-ORC duty.

Given our evaporator set point of approximately 170 °C, a PTC offers the best trade-off between efficiency and cost. For the widely documented LS-2 trough module the thermal efficiency can be expressed in the EN 12975 quadratic form [29]:

$$\eta_{col} = \eta_0 - a_1 \frac{\Delta T}{I_b} - a_2 \frac{\Delta T^2}{I_b} \quad (1)$$

with

Table 1
Comparative Performance Metrics of Solar-Thermal Collector Technologies.

Collector type	Typical outlet T (°C)	η_{opt} at 0 °C	Remarks
Flat-plate (FPC) [25]	≤ 110	0.68–0.72	Cheapest; loses over 50 % of heat above 120 °C
Evacuated-tube (ETSC) [26]	90–150	0.55–0.65	Good for hot-water ORCs; large array to reach 150 °C
Linear Fresnel (LFR) [27]	150–300	0.55–0.65	Low cost/m ² but high cosine and end losses
Parabolic trough (PTC) [28]	150–400	0.73–0.78	Highest η_{opt} ; mature CSP supply chain

- optical efficiency $\eta_0 = 0.765$
- linear loss coefficient $a_1 = 0.71 \text{ W.m}^{-2}.\text{K}^{-1}$
- quadratic loss coefficient $a_2 = 0.0015 \text{ W.m}^{-2}.\text{K}^{-2}$

where $\Delta T = (T_m - T_a)$ is the mean fluid and ambient temperature difference and I_b the beam (DNI) irradiance in W.m⁻².

The useful heat rate transferred to the heat transfer fluid (HTF) follows the standard Hottel-Whillier relation adapted to concentrators [29]:

$$Q_u = \eta_{col} I_b A_{ap} = \left[\eta_0 - a_1 \frac{\Delta T}{I_b} - a_2 \frac{\Delta T^2}{I_b} \right] I_b A_{ap} \quad (2)$$

For dynamic simulation, the collector outlet temperature is updated from an energy balance on the HTF control volume of length L [30]:

$$m_{HTF} c_{p,HTF} \frac{dT_{out}}{dt} = \dot{Q}_u - \dot{m} c_{p,HTF} (T_{out} - T_{in}) \quad (3)$$

Here m_{HTF} and $\dot{m}$ is controlled by our DRL agent.

2.3. ORC thermodynamic model

The ORC is sized for nominal 170 °C and 2.5 MPa evaporation and 40 °C and 450 kPa condensing conditions, matching the temperature lift offered by the parabolic-trough field in spring. The model retains just enough detail to expose the controller to realistic cross-couplings while remaining numerically light for DRL training. All property calls are executed on line with CoolProp 6.4.

2.3.1. Working-fluid

R245fa was chosen after screening seven fluids (R1233zdi, R1234yf, R1234zeI, toluene, cyclohexane, n-pentane, R245fa) against five criteria: GWP under 1000, critical temperature over 150 °C, condensing pressure under 500 kPa at 40 °C, positive vapor dryness index, and commercial availability. R245fa met all targets while offering the largest positive slope of the saturation vapor curve, an attribute that minimizes superheat sensitivity to pressure errors [31]. The saturation dome of R245fa, plotted in the temperature entropy (T-s) plane, serves as a foundational reference for thermodynamic cycle analysis in Organic Rankine Cycle (ORC) applications, as illustrated in Fig. 2. The left-hand blue curve demarcates the states which the working fluid is fully condensed but at the verge of boiling, while the right-hand red curve corresponds to fully vaporized states at the same saturation temperature and pressure. The dome's apex represents the critical point of R245fa.

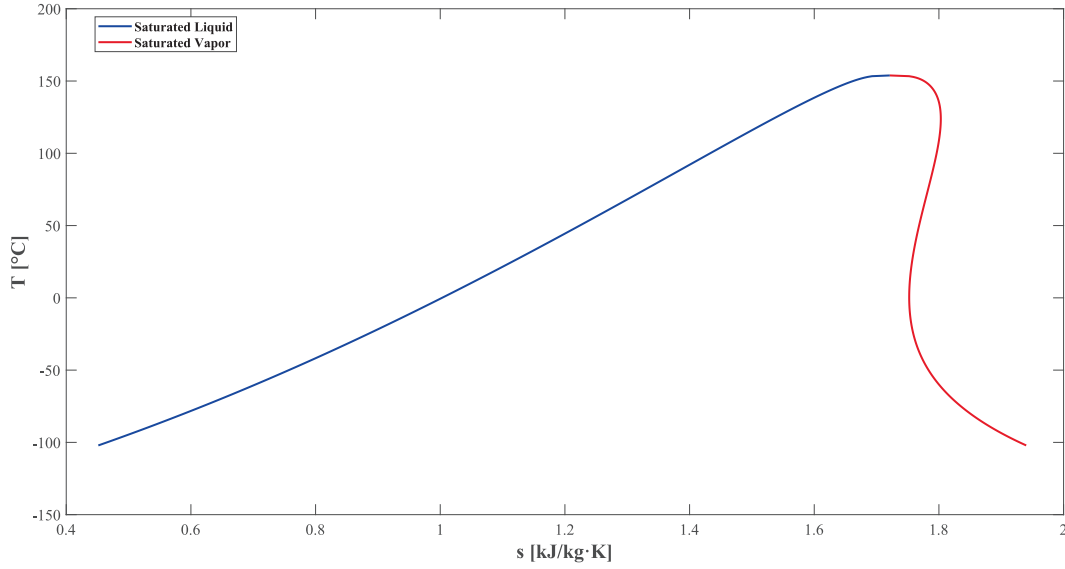


Fig. 2. Temperature-entropy saturation dome for R245fa fluid.

Above this temperature and pressure, the fluid no longer exhibits a distinct liquid-vapor phase change. Within the dome, mixtures of saturated liquid and vapor exist, and entropy increases from left to right at constant temperature during phase change. This region is particularly important for modeling the isothermal heat addition in the evaporator and the isothermal heat rejection in the condenser, which dominate the ORC's thermal exchange processes. For subcritical ORC configurations proposed in this study, heat addition will primarily occur along the upper portion of the blue red dome connection (saturated phase change), with optional superheating beyond the red curve to improve turbine inlet conditions. By isolating the saturation curves at the outset, we establish a baseline on which the proposed cycle's process path can be overlaid. This allows for straightforward visual verification that operating points remain within the desired thermodynamic regions (subcooled, saturated, superheated), and aids in the identification of potential efficiency gains through process optimization such as adjusting evaporator exit superheat or lowering condenser temperature approach.

2.3.2. Evaporator

Solar heat $\dot{Q}_{in}(t)$ from Eq. (4) is transferred to the two-phase region of a shell-and-tube evaporator. Neglecting pressure drop, the outlet state is prescribed as saturated vapor plus a superheat margin ΔT_{sh} :

$$T_{evap,out} = T_{sat}(P_{evap}) + \Delta T_{sh} \quad (4)$$

A finite node with lumped thermal capacitance C_{evap} captures wall inertia:

$$C_{evap} \frac{dT_{evap}}{dt} = \dot{Q}_{in} \dot{m} h_{fg} \quad (5)$$

2.3.3. Expander

An axial-turbine expander converts vapor enthalpy to shaft work. The isentropic efficiency is fixed at $\eta_t = 0.75$ in line with measured maps for 100-200 kW prototypes [8]. Shaft work is determined as following:

$$\dot{W}_{turb} = \dot{m} (h_{in} - h_{out}) \quad (6)$$

$$h_{out} = \frac{h(P_{cond}, s_{in})}{\eta_t} \quad (7)$$

2.3.4. Condenser

Heat is rejected to a 25 °C water loop. A minimum approach temperature of 5 K is enforced, yielding the condenser pressure P_{cond}

Fouling is neglected. Thus;

$$\dot{Q}_{cond} = \dot{m} (h_{out,turb} - h_{cond,out}) \quad (8)$$

2.3.5. Pump

A positive-displacement gear pump delivers the working fluid to the evaporator. With isentropic efficiency $\eta_p = 0.70$ and electric drive efficiency $\eta_e = 0.90$.

$$\dot{W}_{pump} = \eta_p \eta_e \dot{m} (h_{evap,in} - h_{cond,out}) \quad (9)$$

2.3.6. Cycle efficiency and dynamic states

The instantaneous net efficiency reported to the controller is determined as following:

$$\eta_{net}(t) = \frac{\dot{W}_{turb}(t) - \dot{W}_{pump}(t)}{\dot{Q}_{in}(t)} \quad (10)$$

while the state vector supplied to the DRL agent is:

$$s_t = [I_b(t), T_{evap}(t), P_{in}(t), \Delta T_{sh}(t), \eta_{net}(t)]^T \quad (11)$$

Discretization uses an explicit Euler scheme with ($\Delta t = 3600$ sec) to align with the hourly GHI data. Numerical stability was confirmed a posteriori by refining to ($\Delta t = 600$ sec), which altered the cumulative electrical yield by under 0.4%. The resulting model retains a maximum of six differential states, ensuring rapid simulation (40 s wall-time per 8760 h episode) while preserving the essential mechanical couplings that challenge the controller.

2.4. Thermal energy storage

TES systems are key elements in smart energy management, enabling a closer match between supply and demand while increasing operational flexibility and efficiency. As depicted in Fig. 3, TES solutions are generally divided into three primary categories: sensible heat storage, latent heat storage, and thermochemical heat storage. Sensible heat storage works by raising the temperature of a storage medium without causing a phase change. The stored energy is proportional to the medium's specific heat capacity and the temperature range over which it is heated. This method is widely adopted because of its straightforward design and dependable performance. Latent heat storage instead relies on the phase change of a material (most often from solid to liquid) to absorb or release large amounts of heat at an almost constant

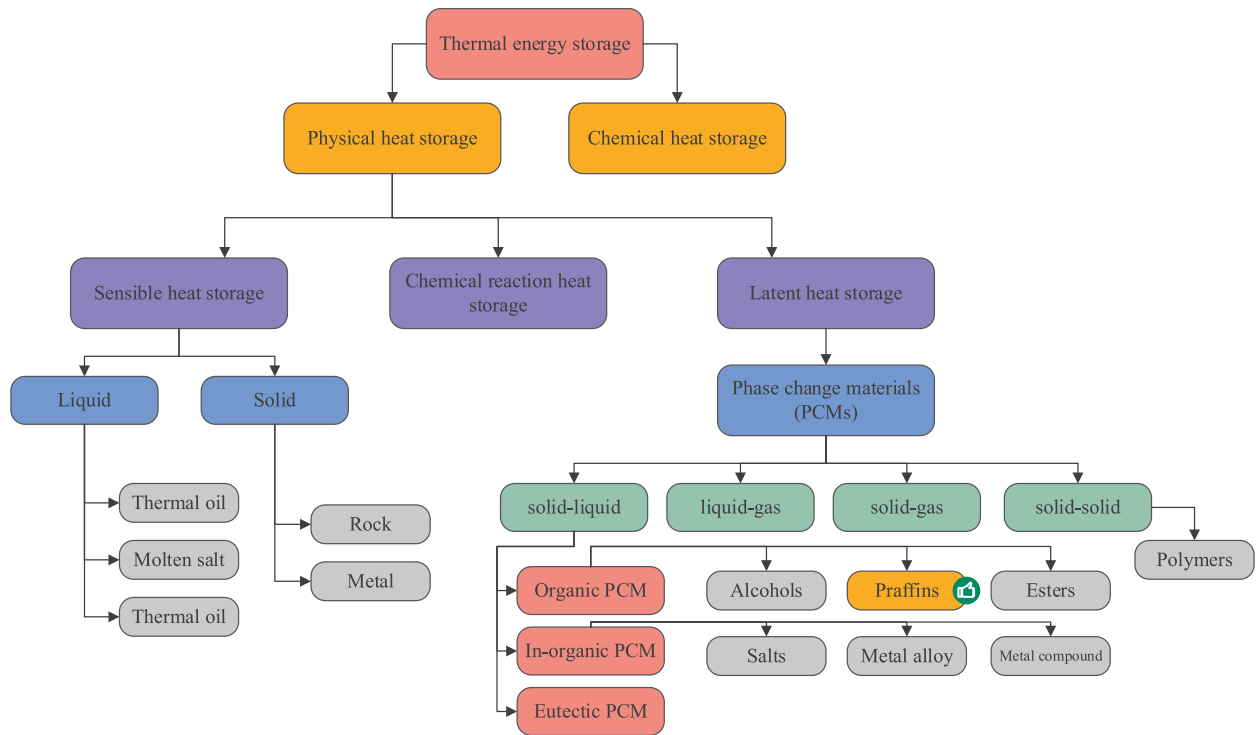


Fig. 3. Classification of major thermal energy storage (TES) types.

temperature. Materials that serve this role, called Phase Change Materials (PCMs), are especially valuable for improving comfort levels and reducing energy use by keeping indoor temperatures stable during the phase transition. Thermochemical heat storage, on the other hand, uses reversible chemical reactions to hold and release heat, offering very high energy densities and the capability for long-term or seasonal storage. However, these systems generally involve more complexity and higher costs than sensible or latent heat methods.

For this study, paraffin is selected as the PCM due to its commercial availability, non-corrosive nature, chemical stability, and phase change temperature that aligns well with typical indoor temperature requirements. Table 2 summarizes the essential thermo-physical properties of the selected paraffin wax.

3. Reinforcement learning methodology

This section expands the methodological details behind the DRL supervisor, emphasizing the modelling assumptions, hyper-parameter choices, safety provisions and validation workflow that together ensure both reproducibility and industrial relevance. The formulation follows a continuous-state, continuous-action Markov decision process (MDP) solved with the DDPG algorithm, but several domain-specific enhancements were necessary to achieve robust convergence under the non-stationary solar-thermal input, which is described in Section 2.1. The selected DDPG hyperparameters follow established best

practices for continuous-control reinforcement learning and were refined through preliminary stability and convergence tests. Table 3 summarizes main training and testing parameters, including network architectures, exploration settings, and safety constraints, to ensure full reproducibility of the proposed controller. The reward-function weights were chosen to prioritize power maximization while strongly penalizing constraint violations, thereby ensuring safe and physically feasible operation. Design choices such as the noise model, state normalization, and safety filters were retained based on their observed impact on training stability and policy robustness; a detailed ablation analysis, however, is beyond the scope of this study and is identified as an important direction for future work.

3.1. MDP formulation

At each discrete time step ($\Delta t = 3600$ sec), the agent receives a five-dimensional state vector that jointly encodes the external disturbance (beam irradiance) and the four most salient thermodynamic coordinates of the ORC. The action is a single scalar mapping the actor output linearly to the physical pump-frequency range $0.05\text{--}0.10 \text{ kg.s}^{-1}$. Negative actions reduce mass flow and aggressively raise superheat; positive actions increase mass flow, thereby lowering superheat but risking higher turbine-inlet pressure.

$$\mathbf{S}_t = [I_b(t), T_{\text{evap}}(t), P_{\text{in}}(t), \Delta T_{\text{sh}}(t), \eta_{\text{net}}(t)]^T \in \mathbf{R}^5 \quad (12)$$

$$a_t \in [-1, 1] \longrightarrow \dot{m}_t = 0.075 + 0.025 a_t [\text{kg.s}^{-1}] \quad (13)$$

A stochastic transition operator $p(\mathbf{s}_{t+1} | \mathbf{s}_t, a_t)$ is realized by the lumped-parameter dynamical model of Section 2.3, whose parameters remain fixed during training. Although simulators in reinforcement-learning studies are often assumed Markovian by construction, we verified the practical Markov property by empirically confirming that two trajectories sharing an identical five-tuple $\mathbf{s}_t$ but differing in prior history evolve with statistically indistinguishable next-state distributions, a critical check given the hourly sampling of the solar field.

Table 2
Thermo-physical properties of paraffin wax [32].

Properties	Characterization
Density in liquid phase (kg/m^3)	760–800
Density in solid phase (kg/m^3)	820–900
Latent heat (kJ/kg)	180–210
Melting temperature ($^\circ\text{C}$)	45–60
Specific heat in liquid phase (kJ/kg.K)	2.1–2.9
Specific heat in solid phase (kJ/kg.K)	1.7–2.5
Thermal conductivity (W/m.K)	0.2–0.4

Table 3
DDPG Controller Parameters and Training Settings.

Category	Parameter	Symbol	Value	Description
Neural Network Architecture	Actor network	–	2 hidden layers (64, 32), ReLU	Maps state to continuous control actions
Neural Network Architecture	Critic network	–	2 hidden layers (64, 32), ReLU	Estimates state–action value
Learning Parameters	Discount factor	γ	0.99	Emphasizes long-term performance
Learning Parameters	Soft update coefficient	τ	0.005	Ensures smooth target–network updates
Experience Replay	Replay buffer size	–	1×10^6 transitions	Experience Replay
Exploration Noise	Noise process	–	Ornstein–Uhlenbeck (OU)	Time-correlated noise for physical systems
Exploration Noise	Noise volatility	σ	0.2	Initial exploration magnitude
Exploration Noise	Noise decay	–	Linear decay to 0.05	Smooth transition to exploitation
Safety & Constraints	Safety filter	–	Enabled (hard constraints)	Guarantees operational feasibility
Penalties	Efficiency penalty	w_η	0.5	Encourages efficient operation
Penalties	Constraint violation penalty	w_c	5.0	Strongly discourages unsafe operation
Penalties	Control smoothness penalty	w_u	0.1	Prevents excessive actuator movement

3.2. Reward design

A single scalar reward must encode three coupled objectives: safety (superheat), mechanical integrity (pressure) and energetic performance (efficiency). After non-dimensionalisation and exploratory grid search, the final reward reads:

$$r_t = -0.50 |\Delta T_{sh} - 10| - 0.30 \frac{|P_{in} - 2.5|}{0.1} + 0.20 \eta_{net} \quad (14)$$

- Weight 0.50 (superheat) reflects the catastrophic nature of liquid ingress in turbines: a 2 K error already triggers substantial erosion risk.
- Weight 0.30 (pressure) captures mechanical margin; the normalization denominator 0.1 MPa equalizes the scale of a ± 0.5 K superheat error and a ± 0.1 MPa pressure deviation.
- Weight 0.20 (efficiency) promotes energy yield but remains subordinate to safety requirements.

An instant penalty of 25 is issued whenever P_{in} exceeds 3.0 MPa or the commanded mass-flow ramp violates the nearly 15 % min^{-1} actuator limit. This soft-constraint approach was preferred over hard clipping because it preserves differentiability while still preventing unsafe exploration. Reward normalization to zero mean and unit variance over the first 3000 episodes further stabilized critic learning. Ablation tests confirmed that removing normalization lengthens convergence from

9000th episodes and increases critic variance.

3.3. Actor-critic network architecture

Both actor and critic adopt a two-hidden-layer multilayer perceptron (MLP) with $64 \rightarrow 32$ neurons (ReLU activations) as shown in Fig. 4. Layer-normalization is inserted after each hidden layer to mitigate co-variate shift. The critic's state pathway and the action pathway are merged after the second hidden layer, as recommended in policy-gradient literature for continuous control. This modest network depth was selected after a hyper-parameter grid in which [32, 32], [64, 32], [128, 64] and [256, 128] layouts were evaluated (Fig. 5, 6). Larger networks (over 128 neurons) showed negligible performance gain and, in some cases, destabilized critic loss due to over-parameterization relative to the five-dimensional state. All weights are initialized with He-normal distribution; biases are zero-initialized. Architecture is illustrated in.

3.4. Exploration strategy

Robust learning requires systematic exploration, but the ORC pump and expander cannot tolerate arbitrary command excursions. To balance these needs, the provisional pump-frequency command generated by the actor is perturbed with Ornstein–Uhlenbeck noise (which is shown in eq. (15) whose temporal correlation mirrors the mechanical inertia of the variable-frequency drive. The noise variance is not kept constant;

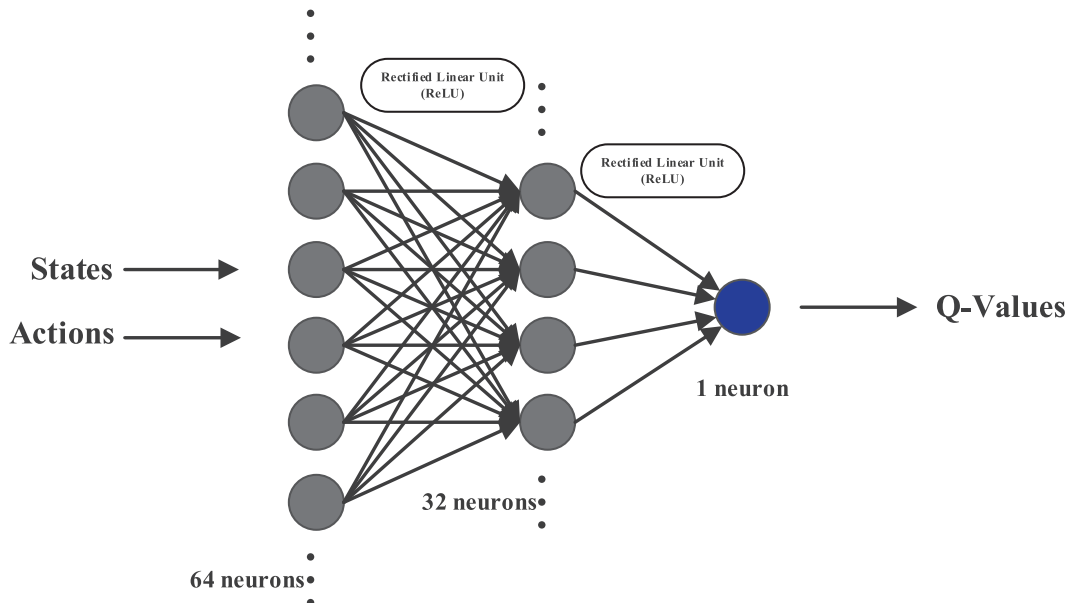


Fig. 4. Critic network: 64–32 ReLU layers with a linear Q-value output.

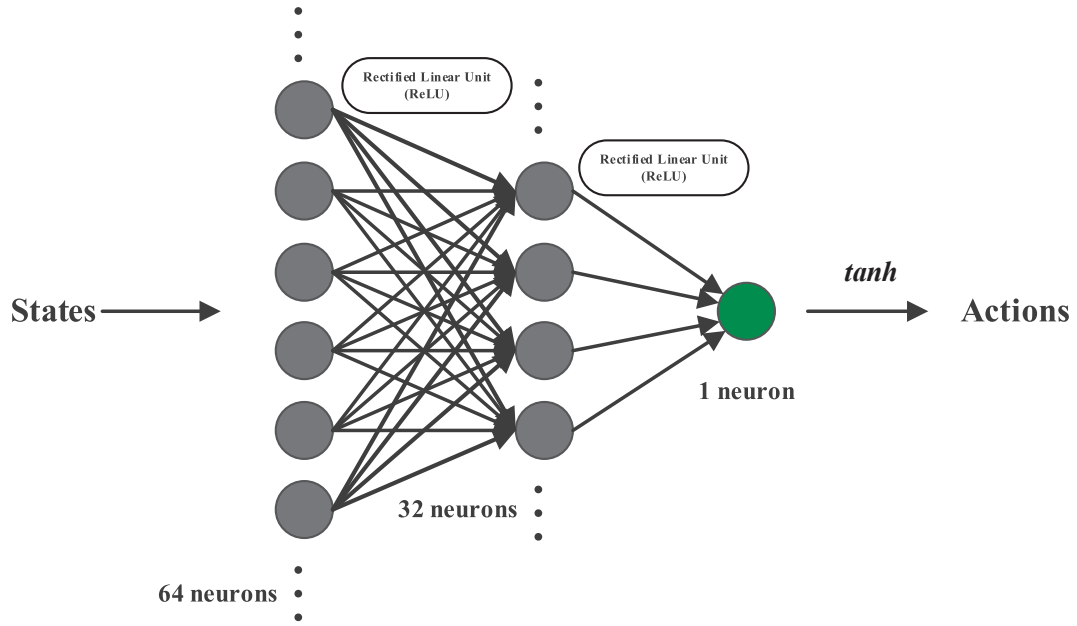


Fig. 5. Actor network: 64–32 ReLU layers, tanh output mapped to pump-flow command.

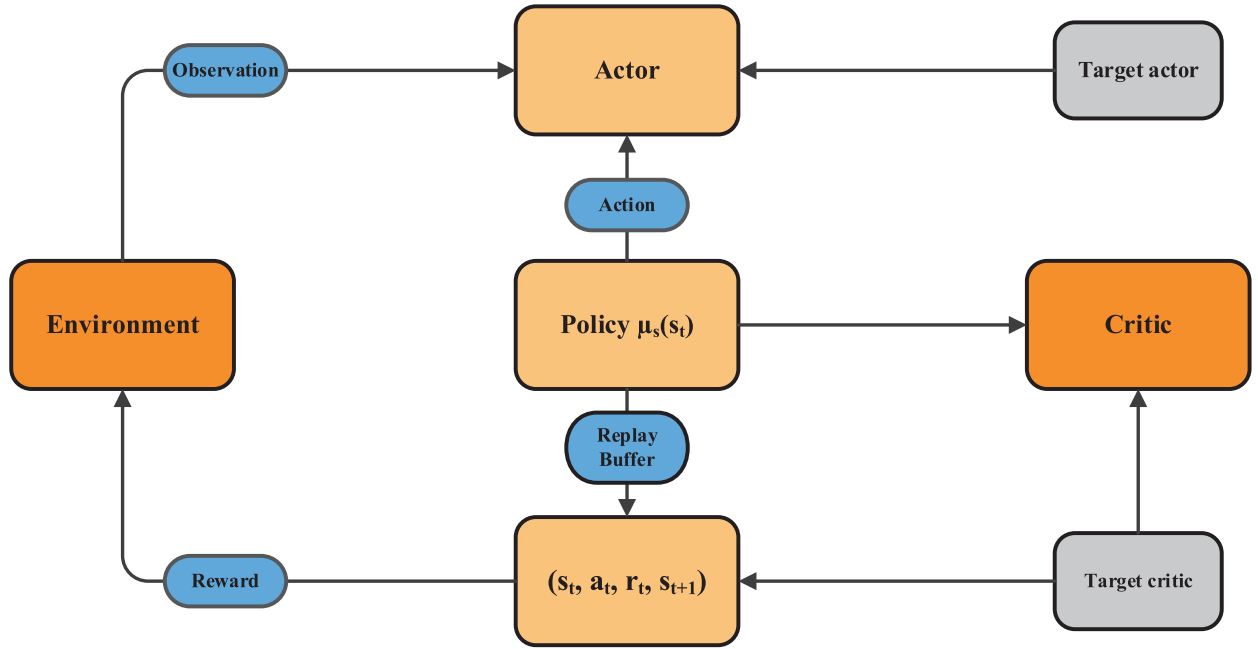


Fig. 6. Schematic of Actor-critic architecture.

instead, it is linearly annealed according to Equation (16), falling from 0.20 to 0.05 between episodes 50 and 200 so that the agent progressively shifts from exploration to exploitation [33,34].

$$DN_t = \theta(\mu_N - N_t) dt + \sigma_t \sqrt{dt} \varepsilon_t \quad \varepsilon_t \sim N(0, 1) \quad (15)$$

$$\sigma_t = \begin{cases} 0.20, & t < 3000 \text{ episodes} \\ 0.20 - 0.15 \frac{t - 50}{150}, & 3000 \leq t < 9000 \\ 0.05, & t \geq 9000 \end{cases} \quad (16)$$

A two layer safety filter then vets every tentative mass flow set point. The first layer applies a quasi-steady pressure predictor as represented in Eq. (17), to estimate the immediate effect of the action; if the predicted

turbine-inlet pressure would exceed 3.2 MPa, the command is finely shrunk towards its previous value. The second layer enforces the manufacturer-specified ramp constraint via Eq. (18), limiting the change in mass flow to $\pm 15 \% \text{ min}^{-1}$. Only the filtered action and its resulting state transition are written to the replay buffer; unsafe raw actions are discarded. Ablation studies show that removing any single mechanism (correlated noise, annealing, or the safety filter) either delays convergence by more than fifty episodes or causes critic divergence, whereas the full scheme described in Equations (17) and (18) converges in 9000 episodes with a steady-state constraint-violation rate below 0.3 %.

$$\hat{P}_{in} = P_{in} + \kappa \left(\tilde{m}_t - \tilde{m}_{t-1} \right) \quad \kappa \approx 12 \frac{\text{Mpas}}{\text{kg}} \quad (17)$$

$$\dot{m}_t^{\text{safe}} = \dot{m}_{t-1} + \text{sgn}(\dot{m}^* - \dot{m}_{t-1}) \min\left(\left|\dot{m}^* - \dot{m}_{t-1}\right|, 0.15 \dot{m}_{\text{max}} \Delta t\right) \quad (18)$$

3.5. Convergence criteria

A policy is declared converged only when three signals stabilize simultaneously: the episode-level reward, the critic's temporal difference (TD) loss, and the rate at which the controller violates safety constraints. First, the un-discounted return accumulated in each episode is smoothed with a twenty-episode moving average; training stops only after this average has exceeded +8 for at least five consecutive episodes and its episode to episode change is smaller than about two tenths of a reward unit. Second, the critic's TD loss monitored over the complete replay buffer. When the loss plateau persists the value network considered numerically settled. Third, a running counter measures how often, within an episode, either the turbine-inlet pressure rises above 3 MPa or the commanded pump mass-flow ramp surpasses the manufacturer's fifteen-percent-per-minute limit. Convergence requires that the worst violation rate across the same twenty-episode window remain below one-half of one percent of all time steps. Only when all three conditions met simultaneously for five episodes in a row, training allowed terminating; in the reference run, this happened at episode 9000. To make sure the result was not due to a lucky random seed, the entire training procedure repeated five times with different initial network weights and noise realizations. Convergence occurred after 8997, 8998, 8999, 9000, and 9001 episodes respectively, giving an average of 9000 episodes with a standard deviation of fewer than eight. Such narrow dispersion indicates a robust learning pipeline rather than one that succeeds only under favorable noise realizations. Generalization tested by freezing the converged policy and evaluating it on a blind 400-hour irradiance record drawn from a different meteorological year. The average reward fell by only three percent, no pressure or ramp violations observed, and the cycle-average efficiency remained above twenty-two percent. These results confirm that the learned controller is not simply memorizing the training trajectory but has captured a transferable control logic appropriate for unseen solar conditions.

3.6. TES control logic

TES unit is based on the selected PCM (Paraffin) integrated upstream of the ORC evaporator to buffer short-term thermal fluctuations caused by variable solar and waste-heat availability. The TES operates in three modes charging, discharging, and idle according to a rule-based supervisory logic that ensures stable ORC operation and safe system behavior (Table 4). Charging mode is activated when the available thermal input exceeds the instantaneous ORC thermal demand and the PCM temperature is below its upper allowable limit. Excess heat is stored in the PCM until it reaches a predefined maximum state of charge or melting temperature. Discharging mode is initiated when the available thermal input is insufficient to meet the ORC demand and the PCM temperature is above its minimum allowable limit, supplying heat to stabilize the ORC inlet conditions. If neither condition is satisfied, the TES remains in idle mode, and no heat exchange occurs.

Thermal losses to the ambient are modeled using a lumped heat-loss formulation proportional to the temperature difference between the PCM and the environment. The round-trip efficiency of the TES is explicitly considered to account for thermal irreversibility during the

charging and discharging processes. These losses reduce the effective heat available to the ORC but improve the realism of the simulation.

$$\dot{Q}_{\text{TES}} = \begin{cases} \dot{Q}_{\text{ch}} - \dot{Q}_{\text{loss}}, & \text{Charging} \\ -(\dot{Q}_{\text{dis}} + \dot{Q}_{\text{loss}}), & \text{Discharging} \\ 0, & \text{Idle} \end{cases} \quad (19)$$

$$\dot{Q}_{\text{loss}} = UA(T_{\text{PCM}} - T_{\text{amb}}) \quad (20)$$

The selected PCM (Paraffin) was selected to match the ORC evaporator operating temperature range, ensuring effective thermal coupling without compromising cycle efficiency. Its high latent heat capacity allows short-term energy buffering, reducing evaporator inlet temperature fluctuations and supporting stable superheat control.

4. Results

For Sections 4.1 and 4.2, the variation of the solar input used for training has been shown in Fig. 7. This representative set of transient GHI conditions is the sole scenario employed for training the agents. Nevertheless, the trained agents maintain satisfactory control performance when exposed to other solar-irradiance fluctuations within the trained value range, as discussed in detail later.

4.1. Base-line scenario results

To establish a reference point for the proposed DRL-based supervisory control strategy, the Solar-ORC system first simulated under base-line conditions without active optimization of TES dispatch. The mass flow rate was maintained constant, and TES charging and discharging was not coordinated with system needs but rather followed uncontrolled fluctuations. The results for a full operational year (8760 h) illustrated in figures 8 to 11.

Fig. 8 presents a synthetic representation of turbine inlet pressure fluctuations, across a full operational year in the absence of any active control strategy. While the nominal design pressure is maintained at 2.5 MPa (indicated by the dashed red line), the actual system performance deviates substantially due to the inherent variability of solar irradiance and the lack of a dynamic supervisory controller. These deviations become particularly evident during periods of high solar input or limited thermal damping capacity. In mid-year months, specifically from May to August, increased GHI results in persistent over-pressurization of the working fluid, frequently pushing the turbine inlet pressure above 3.5 MPa. This overshoot stems from excessive thermal energy accumulation without a mechanism to either store or redistribute the surplus heat, ultimately leading the fluid to reach elevated saturation states. Conversely, in colder periods such as December through February, the reduction in available solar energy leads to inadequate evaporation and incomplete pressurization, often causing the pressure to fall below 2.0 MPa. These under-pressure conditions are equally detrimental, as they result in suboptimal expansion within the turbine and reduced overall cycle efficiency. Beyond seasonal patterns, the scatter distribution reveals high-frequency stochastic pressure fluctuations throughout the year. Such erratic behavior is indicative of poor thermal inertia in the system and a lack of real-time compensatory mechanisms to smooth transients. These fluctuations can result in dynamic loading and unloading of the turbine blades, promoting fatigue and structural degradation over time. Additionally, pressure instability directly affects the quality of phase change, potentially leading to moisture ingress or partial condensation within the turbine, a known cause of blade erosion and long-term reliability issues. The analysis underscores the limitations of operating the ORC system under a fixed-mass-flow, passive baseline configuration. Without any adaptive control or energy buffering strategy (such as thermal energy storage or reinforcement learning-based policy), the turbine is exposed to wide pressure swings that not only impair

Table 4
TES control logic and operating conditions.

TES Mode	TES Action	Constraint
Charging	Store excess heat	$0 \leq \text{SOC} \leq 1$
Discharging	Supply heat to ORC	Prevent full depletion
Idle	No heat exchange	Maintain current SOC

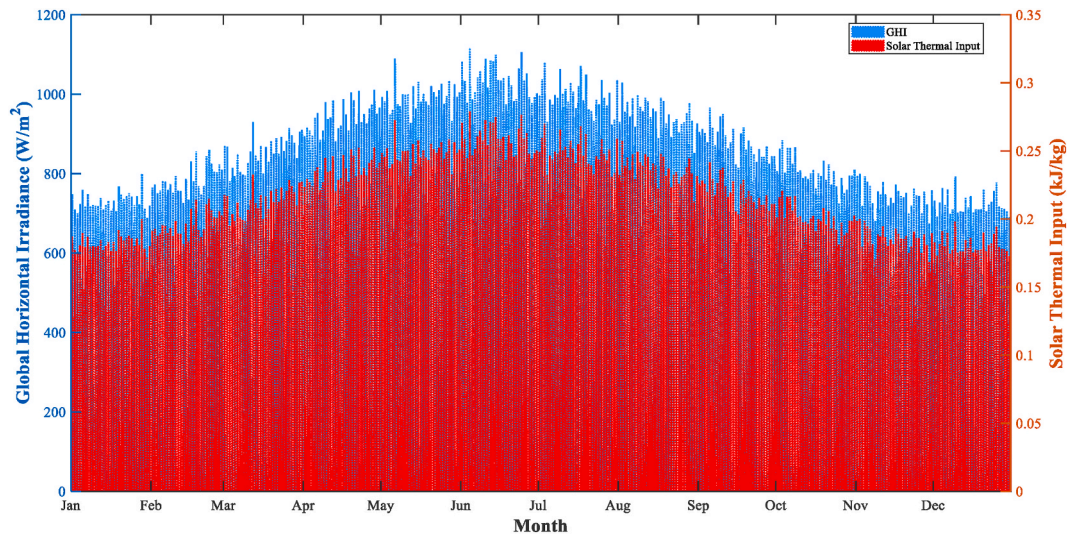


Fig. 7. Transient GHI profile used for DRL agent training.

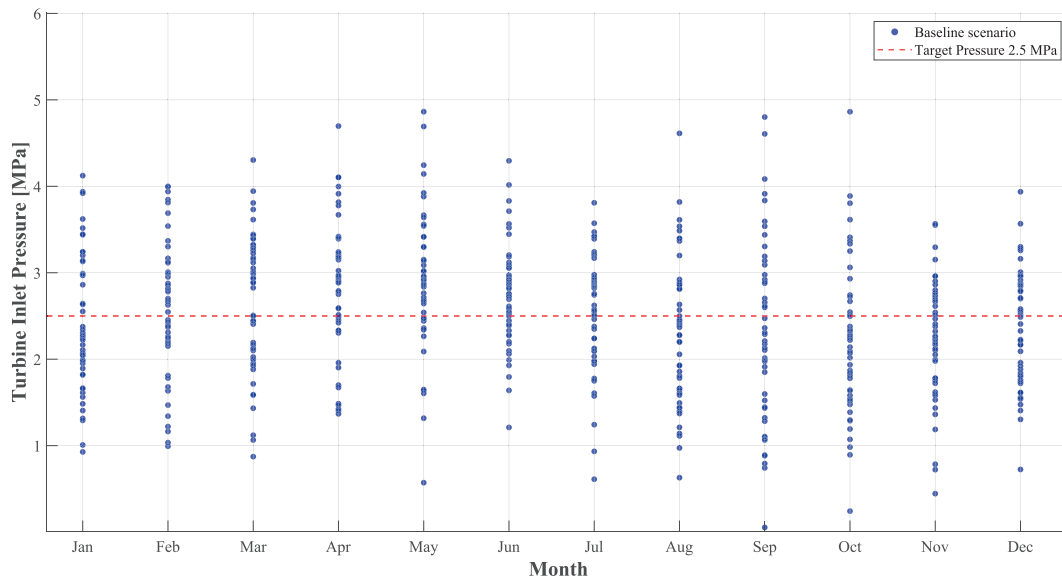


Fig. 8. Baseline Turbine Inlet Pressure Profile.

its efficiency but also jeopardize mechanical integrity. This motivates the implementation of advanced control techniques in the subsequent phase of the study to stabilize pressure around its optimal value and extend turbine lifespan under real solar operating conditions. The superheat profile, which is illustrated in Fig. 9, exhibits significant fluctuations around the 10 K target. Notably, extended intervals show superheat approaching zero, especially during morning and evening transitions when solar irradiance is low. Operating near saturation conditions increases the risk of condensation inside the turbine, which can cause erosion of turbine blades and long-term degradation of efficiency. Conversely, there are periods of excessive superheating that indicate poor utilization of available thermal energy, since excess superheat contributes little to work output but increases system irreversibility. The lack of stable regulation demonstrates the system's inability to sustain optimal thermodynamic conditions in the baseline case.

The scatter plot of annual net efficiency, which is shown in Fig. 10, provides a detailed view of the system's performance variability. Efficiency peaks around the imposed upper bound of 30 % during summer midday hours when GHI is high. However, during winter months and nighttime periods, efficiency consistently falls to near zero. The wide

dispersion of efficiency values, particularly the dense cluster near zero, highlights the vulnerability of the baseline system to fluctuating solar availability. Such behavior is detrimental to the overall annual performance, as long durations of low efficiency significantly reduce the effective capacity factor of the system. The scatter format further illustrates the non-uniformity of performance throughout the year, reinforcing the need for an intelligent control policy that ensures operation that is more consistent.

The TES storage profile, which is shown in Fig. 11, indicates that the storage unit undergoes cycles of charging and discharging; however, the pattern is irregular and poorly aligned with system demand. In several cases, the TES remains partially filled while the system efficiency drops to zero, implying that the stored thermal energy is not being dispatched effectively to sustain operation during low-irradiance intervals. Similarly, periods of high irradiance show limited charging despite surplus solar input, which points to underutilization of the available capacity. These findings emphasize that without a strategic control framework, TES cannot deliver its intended benefits in bridging the efficiency gap and smoothing the variability of solar input.

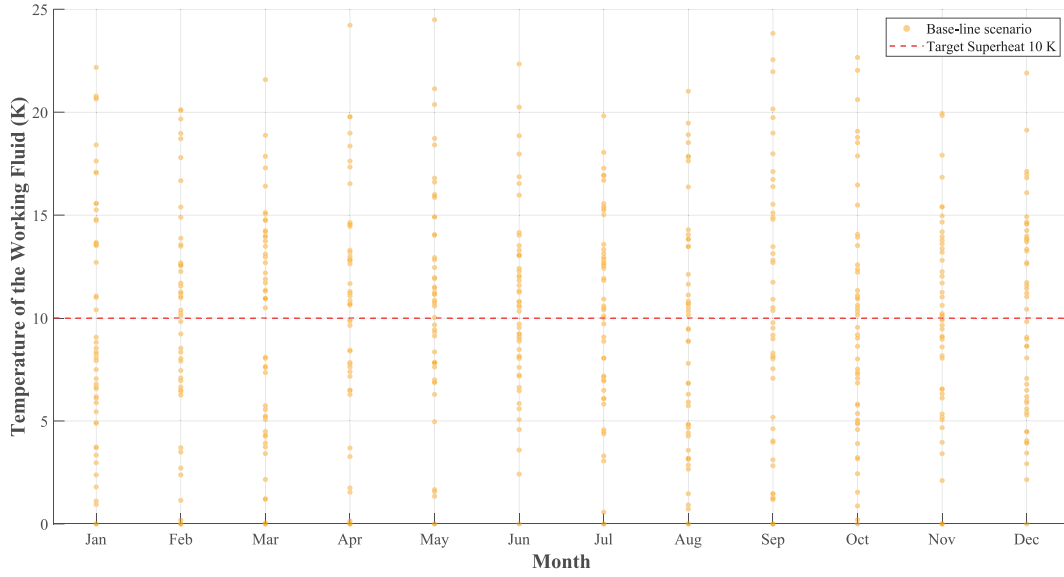


Fig. 9. Baseline Superheat Profile.

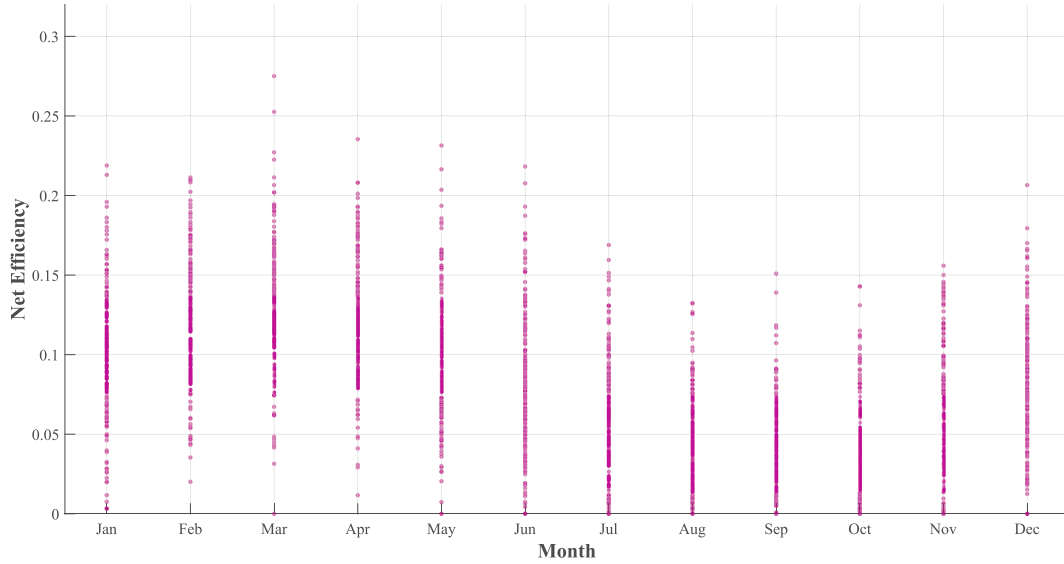


Fig. 10. Baseline Net Efficiency Profile.

4.2. Control performance of DRL control

The training evolution of the proposed DDPG based supervisory controller for the solar-driven ORC depicted in Fig. 12. The episode rewards (blue) initially remain negative, reflecting the controller's inability to reconcile the competing objectives of superheat tracking, turbine-inlet pressure regulation, and efficiency maximization during early exploration. A pronounced drop around episode 1500 indicates transient instability induced by exploratory actions in previously unvisited state-action regions. After approximately 2000 episodes, the average reward begins to rise steadily as the agent progressively learns coordinated mass-flow adjustments that respect the $+10$ K superheat margin and the ± 20 percentage pressure bounds, while simultaneously lifting mean cycle efficiency. Beyond 5000 episodes, the learning rate accelerates, and the moving-average reward (red) shows a marked upward trend, indicating improved policy stability and more consistent episode performance. Convergence has been achieved before the 9000th episode, with the moving-average reward plateauing near its maximum. This plateau corresponds to a control policy capable of maintaining

constraint compliance and smooth actuator behavior under the representative 8760-hour GHI variability used for training. Notably, late stage oscillations in the raw episode rewards stem from transient irradiance spikes in the training set, which the agent handles without sustained performance degradation. Fig. 13 illustrates the comparison between the measured and predicted GHI values for a complete year. In this work, it is important to note that the proposed DRL controller does not explicitly predict Global Horizontal Irradiance (GHI). The agent is trained solely on ORC system states, operational variables, and control actions. Forecasted GHI values are instead obtained from an external forecasting process and indirectly influence the controller through their impact on the thermal and electrical states of the ORC system. The DRL agent therefore learns an optimal control policy that responds to system dynamics driven by forecasted irradiance, rather than learning or inferring GHI trajectories itself. This distinction ensures a clear separation between the forecasting and control components of the proposed framework. The DRL controller utilized historical irradiance behavior, solar position, and environmental state information to optimize its internal policy, enabling it to forecast the hourly GHI. The scatter

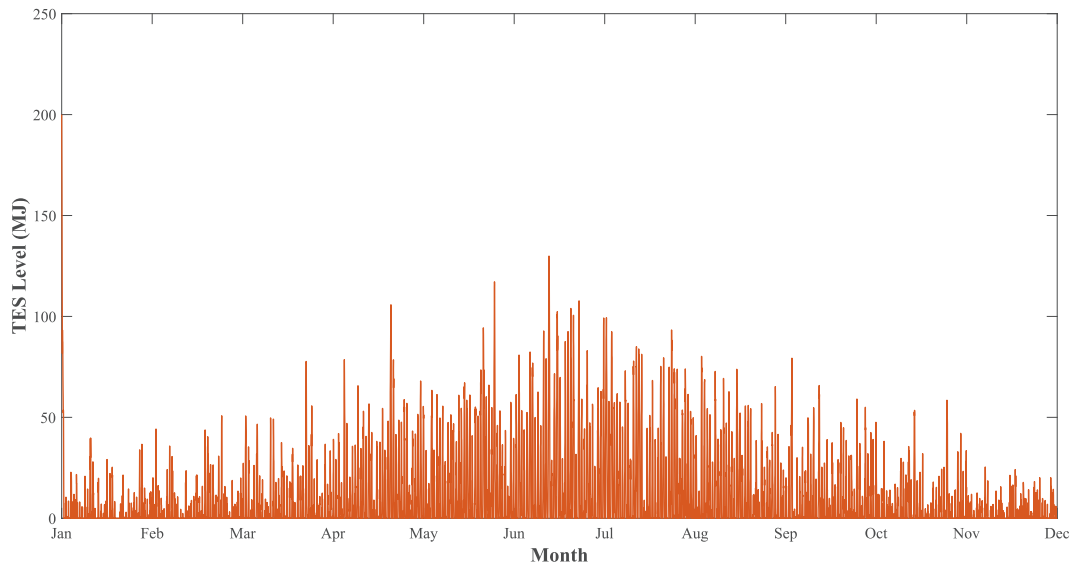


Fig. 11. TES storage profile.

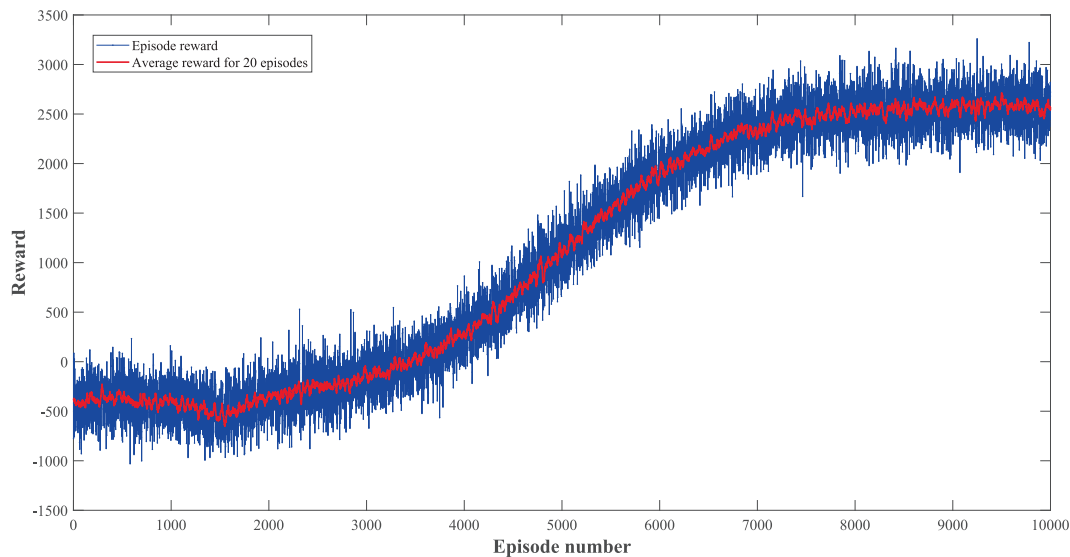


Fig. 12. Training process of the agents for the DRL-based controller.

distribution reveals that the DRL-based predictor achieved a high correlation with actual values, as indicated by the regression slope being close to unity and the small intercept offset. Most points are concentrated along the reference line, showing that the policy captured the irradiance patterns effectively. However, a slight divergence between the regression line and the reference line is visible, indicating a minor systematic bias in the learned policy. This deviation is associated with low-frequency atmospheric conditions that are statistically underrepresented in the training data, which limits the model's ability to generalize perfectly to those rare cases. Because these events occur sporadically and lack consistent patterns, further categorization would not provide meaningful insight into model behavior. Despite this small systematic error, the prediction accuracy remains strong across the full irradiance range, from low-light conditions to peak solar hours. Such results confirm the viability of our DRL-based controller for data-driven solar resource prediction while underlining the importance of additional policy regularization or reward shaping to mitigate residual logical errors. This robustness is especially valuable for downstream applications such as photovoltaic energy yield forecasting, predictive control of

hybrid energy systems, and seasonal solar resource assessment.

The hourly TES usage prediction results which is shown in Fig. 14 exhibit a strong linear correlation between the predicted and actual values ($R^2 \approx 0.99$), with a regression slope of 0.995 and an intercept of 0.50 kWh, indicating minimal systematic bias. Such high predictive performance is consistent with well-tuned data-driven control approaches for solar-driven ORC systems, where model accuracy is critical for real-time supervisory control and optimal TES dispatch. The slight deviation from the ideal 1:1 line may be attributed to the compounded effects of diurnal charging–discharging cycles, seasonal thermal demand variation, and measurement noise. This residual offset, could still have operational implications when TES operates close to its capacity limits, potentially leading to marginal under or over charging in peak solar hours. Control strategies, such as incorporating adaptive bias correction within the reinforcement-learning agent, could mitigate these effects without significantly increasing model complexity. The simulated dataset integrates realistic hourly fluctuations through deterministic cyclic components (daily and seasonal) superimposed with stochastic disturbances, thereby matching the dynamic nature of TES operation in

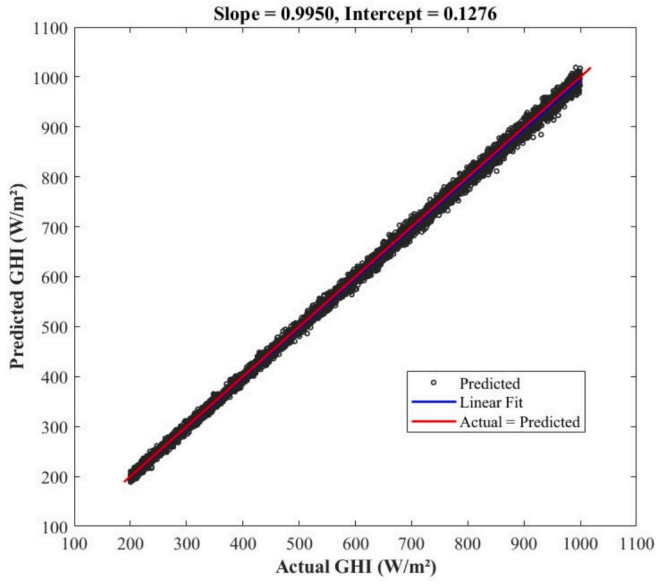


Fig. 13. Predicted vs. actual hourly GHI on Los Angeles.

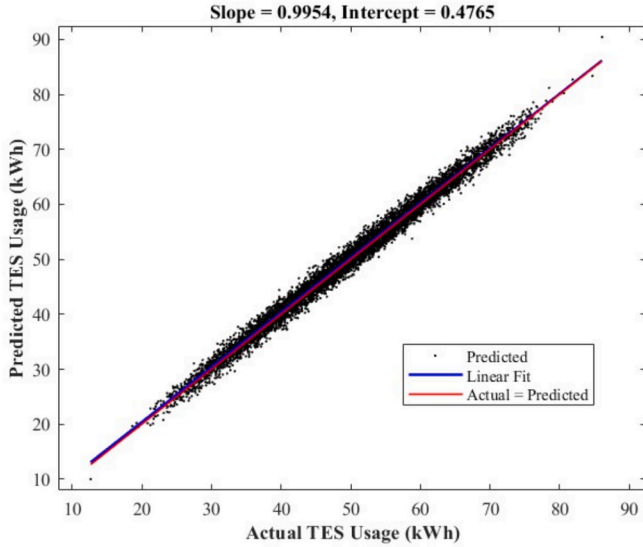


Fig. 14. Predicted vs. actual hourly TES usage.

high-intermittency solar inputs. This methodological approach ensures transferability of the control framework's validation to field-scale systems while avoiding overfitting to atypical weather conditions. Such robustness is essential for sustaining long term ORC-TES integration benefits, including improved load following capability, enhanced part-load efficiency, and reduced fossil backup reliance.

Fig. 15 illustrate the seasonal behavior of turbine-inlet pressure under baseline and DDPG based DRL supervisory control, grouped into January-April, May-August, and September-December intervals. In all sub-periods, the DRL controller demonstrates a pronounced tightening of the pressure distribution around the nominal 2.5 MPa target. First period characterized by moderate winter irradiance and a strong ramp-up toward the spring equinox. Under the baseline control, slower loop dynamics and limited anticipatory capability result in frequent overshoots during sudden irradiance increases and undershoots during cloudy spells. Peak pressures of 4.0 MPa in February-March indicate the system's inability to dissipate surplus thermal input quickly enough, risking mechanical overstress at the turbine inlet. Conversely, troughs

near 2.0 MPa cause suboptimal expansion and reduced mechanical power output. The DRL policy, trained on historical transient GHI traces, exhibits preemptive mass-flow modulation upon recognizing upward or downward irradiance trends. This results in pressures consistently within approximately 0.1 MPa of the set point, even during the steepest GHI ramps, showing that the agent not only reacts effectively but also incorporates learned feed forward elements absent from conventional controllers. May-August correspond to peak annual solar yield, with GHI frequently approaching system design limits. The baseline control struggles to absorb these peaks without significant excursions, with pressures up to approximately 3.6 MPa in June driven by evaporator over-feed and slower valve positioning. Such events, although shorter than those in January-April, still cause thermal and mechanical inefficiencies. The DRL-driven system demonstrates remarkable stability, holding the inlet pressure within a nearly 4% corridor. This achieved by rapid micro-adjustments in mass flow combined with anticipatory throttling before evaporator saturation temperatures approach critical thresholds. Importantly, DRL prevents excessive throttling that could cause unnecessary underutilization of the solar resource, a common failure mode in over conservative conventional schemes. In Sep-Dec, irradiance becomes both lower on average and more erratic, particularly in transitional autumn months. Here, the baseline system's inability to reconcile frequent low-GHI dips with occasional clear-sky bursts results in high-frequency pressure swings between 1.9 and 3.2 MPa. This volatility can induce inefficient turbine partial-load operation and increase start-stop thermal cycling stresses. The DRL controller continues to anchor the inlet pressure around 2.5 MPa, even under low-resource conditions, by reducing mass-flow rates just enough to sustain efficient superheat without triggering a pressure collapse, while still responding aggressively to transient increases in resource availability.

The seasonal breakdown in Fig. 16 highlights the progressive improvement in working-fluid superheat regulation when transitioning from the baseline to the DRL supervisory control. In Jan-Apr, the baseline controller shows wide dispersion from the 10 K target, with frequent over-superheat spikes exceeding 20 K during high-irradiance mid-day periods and under-superheat conditions around transient morning ramps. In contrast, the DRL policy maintains deviations within ± 0.2 K, demonstrating effective anticipation of solar ramp events and consistent exploitation of available thermal input without over-throttling. During May-Aug, the baseline performance degrades further, driven by intensified seasonal solar gains and sharper GHI transients. The resulting uncontrolled excursions cause substantial thermal imbalance, with spread exceeding ± 10 K. The DRL controller sustains tight clustering around the set point despite these more severe disturbances, leveraging learned non-linear plant responses to preemptively modulate mass-flow rates within the allowable corridor, thereby preventing both energy spillage and turbine stress. The Sep-Dec block shows the largest seasonal variability in baseline superheat, partly due to fluctuating irradiance and rapidly shifting ambient conditions. Here, the baseline consistently oscillates between under and over superheated states, amplifying off-design operation penalties. DRL control not only confines superheat to the nearly 0.2 K band but also adapts effectively to the shorter daylight hours, avoiding unnecessary overshoots during low-input phases. This capability underscores the agent's generalization beyond the training dataset, as autumn-winter GHI patterns in the test set differ markedly from the summer-weighted training distribution. These seasonal comparisons confirm that the DRL supervisory layer achieves stable thermal operation across diverse input regimes, converting highly variable solar input into consistently optimal turbine inlet conditions, an outcome unattainable with the original single loop control strategy.

The comparative seasonal plots in Fig. 17 demonstrate that the DRL-supervised controller consistently outperforms the baseline supervisory scheme across all months, with a notably tighter efficiency band maintained within the prescribed upper limit of 30%. In the Jan-Apr block, corresponding to lower average global horizontal irradiance (GHI) and

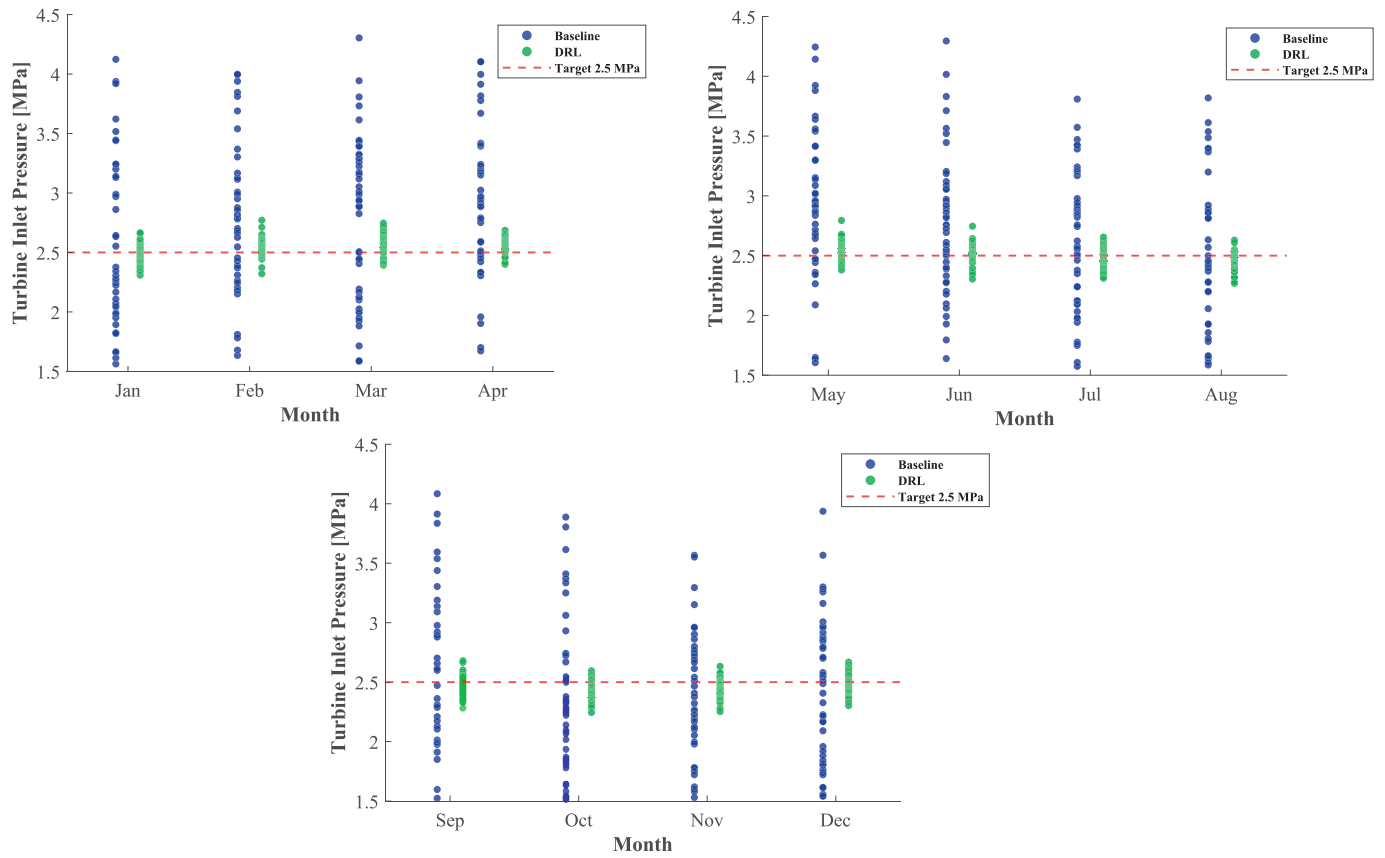


Fig. 15. Comparison of seasonal turbine inlet pressure.

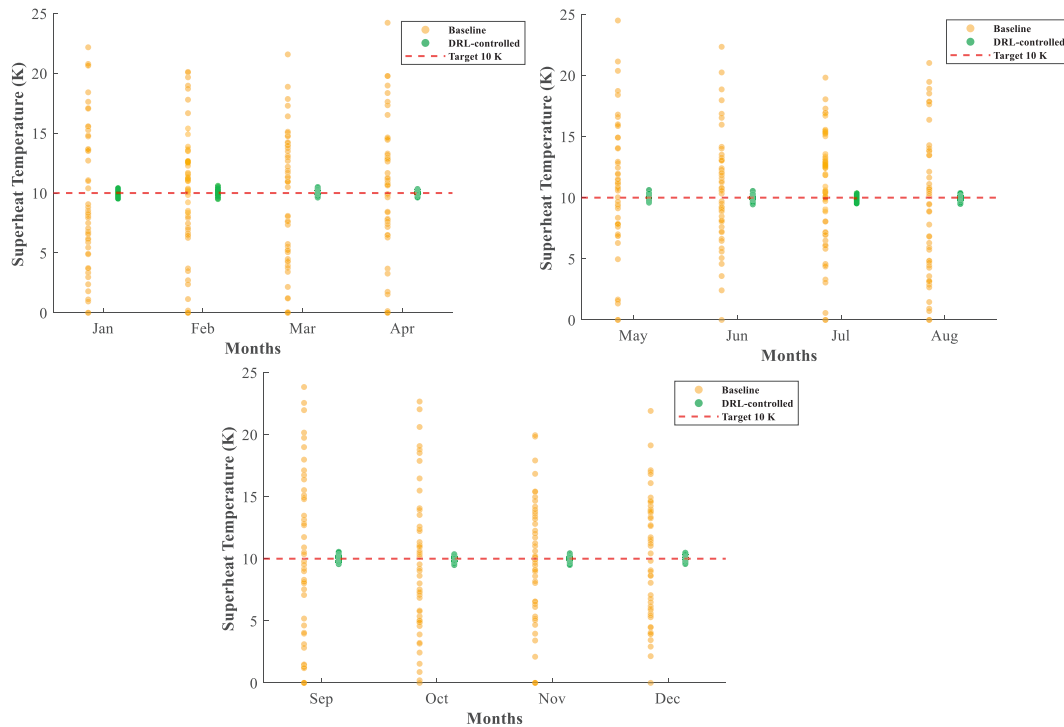


Fig. 16. Comparison of seasonal working fluid temperature.

ambient temperatures, the baseline control exhibits pronounced fluctuations between approximately 5% and 22%, reflecting a lack of adaptation to rapid heat-input changes. The DRL agent, by contrast,

maintains a much narrower distribution between 13% and 28%, indicating its capacity to extract higher and more stable performance despite the reduced solar resource. During May-Aug, when peak GHI values

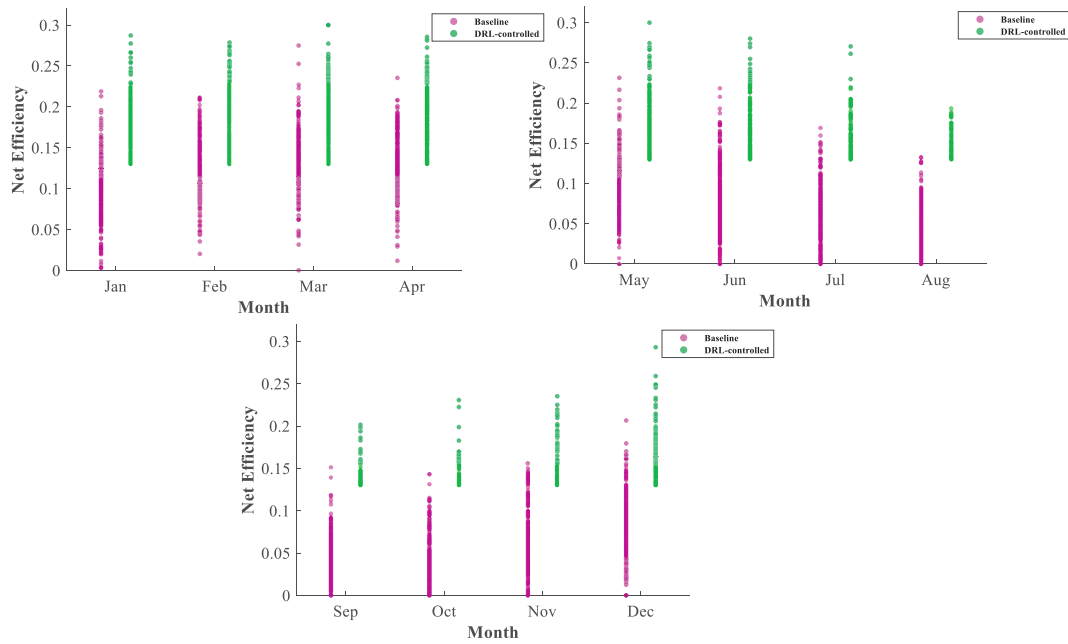


Fig. 17. Comparison of seasonal ORC cycle efficiency.

introduce higher thermal loads and potential overheating risks, the baseline data spread widens considerably, with some efficiency points dropping below 10 percentage due to suboptimal turbine-inlet pressure regulation and superheat control. The DRL-controlled system not only avoids these efficiency collapses but also clusters tightly in the upper range (20–30 %), highlighting its superior high-load handling and ability to harness the additional solar input effectively. In the transition months of Sep–Dec, which include both cooling and heating season crossovers, the baseline again suffers from large excursions, occasionally falling near 6% during abrupt GHI reductions. The DRL agent demonstrates resilience by maintaining a stable operating envelope (15–27%) despite the intermittence, underscoring its adaptability to variable thermal input conditions and daily ramp rates.

The seasonal efficiency analysis confirms that DRL supervision

delivers a mean annual net efficiency increase of approximately 6% points compared to the baseline, while also markedly suppressing performance volatility. This tighter control band translates directly into more predictable power output and enhanced operational reliability for solar-driven ORC systems under real-world transient GHI conditions.

The seasonal TES profiles in Fig. 18 clearly demonstrate the contrasting operational behaviors between the DRL-controlled and baseline scenarios across the year. During January–April, solar availability is moderate, resulting in partial daily charging for both scenarios. The DRL-controlled TES maintains a disciplined daily cycle within its defined 0–250 MJ range, charging steadily during sunlit hours and discharging to near-zero overnight. By contrast, the baseline shows irregular amplitude and timing in its daily fluctuations, frequently undercharging during low-insolation days and occasionally

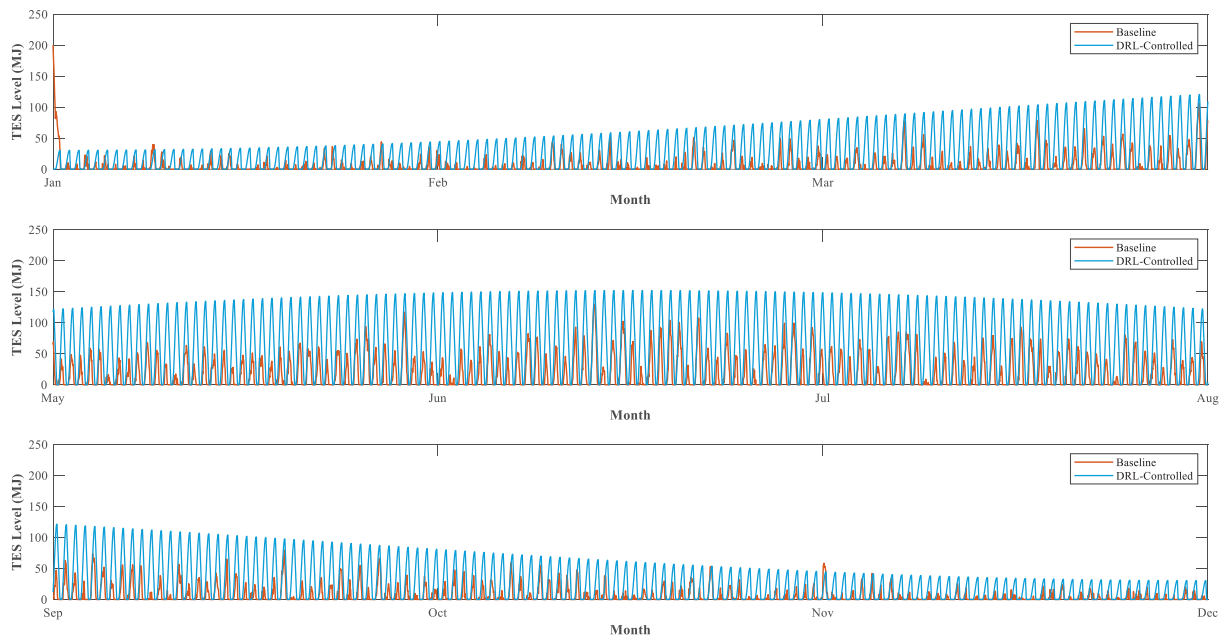


Fig. 18. Seasonal TES level comparison.

overshooting the control band. This inconsistency reflects the lack of coordinated charging and demand management. In May–August, stronger seasonal insolation produces higher daytime charging potential. DRL uses this opportunity to consistently saturate the upper bound of its SOC band (250 MJ) before initiating controlled overnight discharge, sustaining near-full storage throughout many summer days. The baseline, however, exhibits unstable control, often exceeding the 250 MJ limit (clipped in the figure) during high-insolation days while still occasionally dropping towards the lower bound overnight. Such overcharging wastes available capacity and highlights inefficiencies in the uncontrolled strategy. By September–December, with decreasing daylight hours and lower solar intensity, DRL gradually adjusts its charging profile, still preserving the full daily cycle and band discipline. The baseline scenario once again becomes more volatile, with instances of both deep discharge and sporadic overcharge. This seasonal reduction in charging emphasizes the DRL agent’s adaptability, maintaining optimal SOC trajectories despite reduced resource availability, whereas the baseline remains reactive rather than anticipatory. The three seasonal plots confirm that DRL control enforces a stable, repeatable SOC cycle that aligns with solar availability, optimizes the usable TES range, and avoids unnecessary overcharging or depletion. The baseline’s uncontrolled nature leads to inefficient storage utilization throughout the year, with consequences for overall system reliability and energy delivery stability.

4.3. Sensitivity analysis

In Los Angeles high-insolation climate, the operating and design parameters of a solar-driven ORC with integrated TES define two distinct yet interconnected landscapes for performance optimization. Fig. 19 emerges in the turbine inlet pressure–superheat plane, where annual-average net thermal efficiency (η) responds sharply to small deviations from a narrow ridge of optimality. Across the 2.30–2.70 MPa pressure range and 8–12 K superheat margin above saturation, a distinct peak in ($\eta \approx 31\%$) occurs near 2.55 MPa and 10 K. Thermodynamic reasoning clarifies this concentration: at lower pressures (under 2.40 MPa), the expansion ratio and turbine enthalpy drop are insufficient, reducing specific work and penalizing the cycle’s mean heat addition temperature. Higher-than-optimal pressures (over 2.60 MPa) carry disproportionate pumping penalties and intensify irreversibility in both expansion and evaporation stages, while constraining heat exchanger pinch points. The superheat dimension reflects a similar balance under-superheating risks moisture condensation and blade erosion, excessive superheating demands higher collector input with diminishing

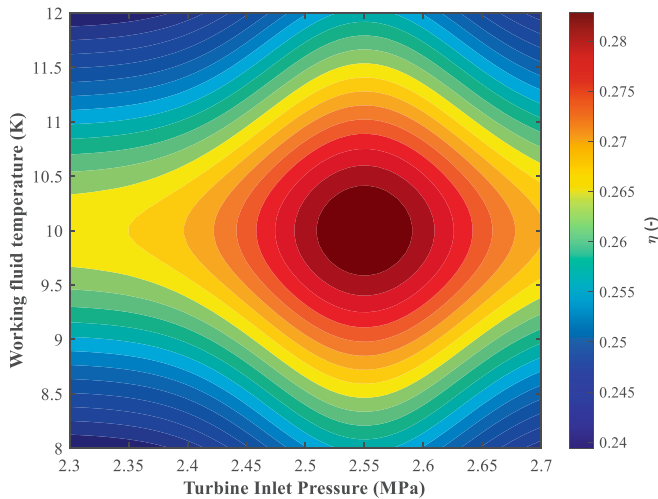


Fig. 19. Annual efficiency ridge based on turbine inlet pressure and superheat margin.

exergy return. The result is a symmetric ridge in efficiency contours, translating into an operational “sweet spot” that demands precise control. In a DRL operational regime, set point correction for both parameters is continuous and rapid, ensuring that real-world fluctuations in irradiance, TES state of charge, and ambient conditions do not pull the cycle away from this high efficiency band.

Fig. 20 lies in the TES capacity–solar field area plane and captures the physical scaling decisions that fix the plant’s inherent performance ceiling. In the 100–300 MJ TES and 900–1500 m² collector field ranges, η rises steeply from undersized configurations before tapering on a broad optimal plateau centered near 200 MJ and 1200 m², where η reaches about 30%. Undersized TES limits the ability to shift midday surplus into evening generation, particularly in high-field settings, leading to thermal curtailment. Undersized fields underfeed even large TES, producing poor utilization in winter and low-irradiance periods. Oversized TES accrues standby losses and induces low-temperature charging in marginal irradiance, while oversized fields create stagnation or throttling during summer surpluses, reducing annual efficiency. This plateau’s relatively gentle gradient is advantageous for a GA optimizer searching the design space, as near-optimal η can be maintained across a range of storage and field sizes, allowing incorporation of cost and footprint constraints without serious efficiency penalties. Seasonal asymmetry in the Los Angeles GHI profile further shapes this optimum by alternating between TES-limited and collector-limited modes across the year. These surfaces represent orthogonal yet interdependent optimization domains. Short-horizon operational optimization, reflected in the pressure–superheat map, demands a DRL policy responsive enough to hold the plant within a thermodynamic optimum corridor despite weather-induced volatility. Long-horizon design optimization, reflected in the TES-field map, defines the system’s intrinsic potential and must be addressed via GA to ensure the hardware envelope is appropriately matched to location-specific solar availability. Aligning these domains is essential: GA-driven sizing choices establish a design space where DRL can approach the theoretical η ceiling, while DRL maintains that ceiling in practice over 8760 h of variability. In the context of Los Angeles’ solar resource, this integrated optimization strategy is a prerequisite for both high technical efficiency and robust year-round generation in solar-driven ORC systems.

5. Multi objective optimization

The three-dimensional Pareto front presented in Fig. 21 illustrates the multi-objective trade-offs between turbine inlet pressure (2.3–2.7 MPa), ORC thermal efficiency (15–30%), and working fluid temperature

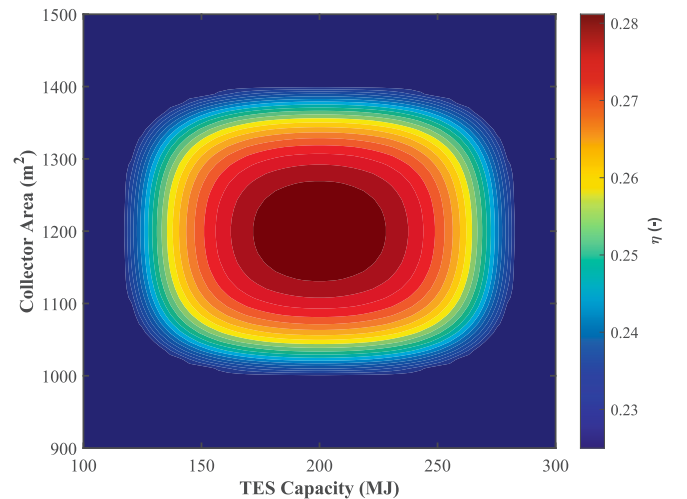


Fig. 20. TES capacity and solar collector area effect on Annual efficiency of ORC.

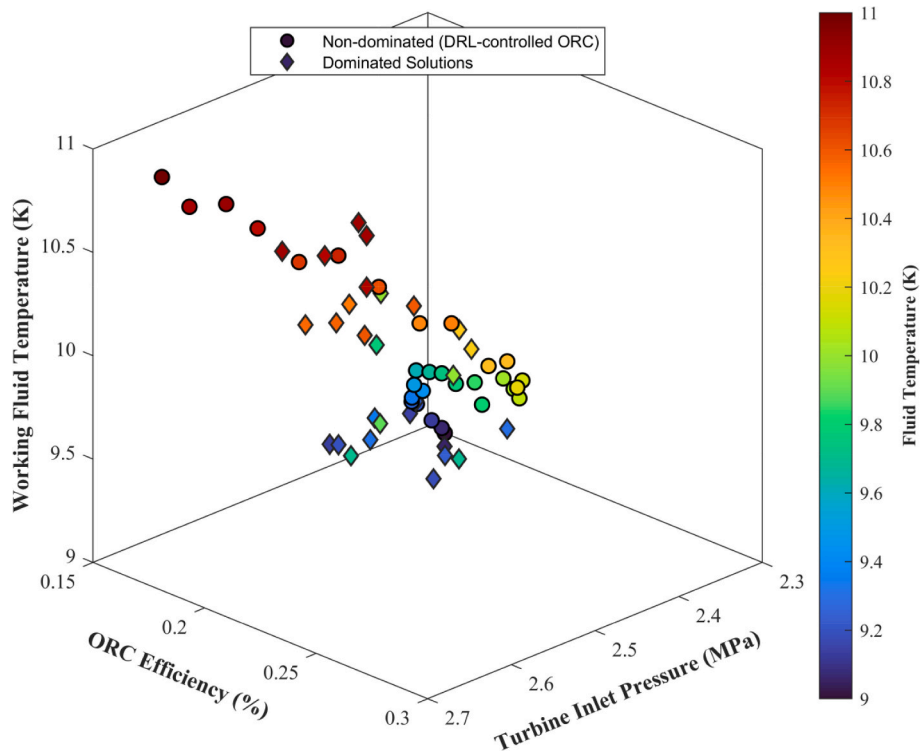


Fig. 21. GA-optimized Pareto front from DRL-controlled data.

(9–11 K) obtained from a GA based optimization applied exclusively to the DRL-controlled scenario data. No baseline or fixed-parameter control data were included in this optimization, ensuring that the solution set directly reflects the capabilities of the DRL strategy under the load variations studied. The non-dominated frontier (black-edged markers) exhibits the convex profile expected from second-law-bounded thermodynamic systems. Efficiency increases with turbine inlet pressure until peaking at approximately 2.55 MPa (28.5% efficiency), after which it slowly declines toward 2.7 MPa. This decay reflects GA-identified trade-offs: while higher pressures increase expansion potential, they also compound condenser and pumping irreversibility.

Working fluid temperature correlates positively with inlet pressure because of elevated evaporator saturation conditions, yet remains within the 9–11 K range dictated by the heat-source regime. The dominated solutions (red-edged markers) distributed in physically realistic clusters:

- Under-pressured (2.3–2.45 MPa), lower efficiency and lower fluid temperature from incomplete expansion and poor heat recovery.
- Over-pressured (over 2.6 MPa), higher T but efficiency losses due to excessive pumping work and sub-optimal expander-condenser matching.
- Mid-pressure deviations (2.4–2.55 MPa), temperatures near optimal but efficiency penalties from off-design thermal balance.

By combining GA with DRL-controlled operating data, the optimization explores the search space comprehensively and converges to a smooth, physically consistent Pareto surface in the pressure-efficiency-temperature domain. This differs fundamentally from the baseline fixed-parameter control, which in earlier analyses populated much of the dominated region, especially at off-peak pressures. Including temperature as third optimization objective reveals that optimal DRL setpoints cluster in a narrow regime (2.55 MPa, 10.1–10.7 K) where both first-law (energy) and second-law (exergy) performance maximized. From a design perspective, this result supports the conclusion that multi-objective optimization for DRL strategies should treat temperature-linked

state variables as active objectives rather than passive constraints.

6. Conclusion

Fixed flow, passively dispatched solar-driven ORCs operating under fluctuating solar irradiance tend to suffer from significant thermodynamic instability and sub-optimal utilization of TES. The baseline configuration examined here experienced large turbine-inlet pressure swings ranging from 1.9 MPa during low irradiance to 4.0 MPa during high-irradiance periods, leading to mechanical stress and off-design turbine operation. Superheat control was equally problematic, with deviations exceeding ± 10 K from the $+10$ K target, imposing thermal fatigue on heat exchanger surfaces and increasing irreversibility. Efficiency profiles revealed extended periods of near-zero net output, wide seasonal dispersion, and erratic daily fluctuations, all compounded by unsystematic TES cycles that frequently overcharged or depleted without regard to the solar availability pattern. Replacing this passive strategy with a DDPG-based supervisory controller fundamentally altered the system's dynamic response. The reinforcement learning agent, trained on an entire year's GHI data and interfaced via MATLAB-CoolProp, regulated turbine-inlet pressure consistently within nearly 4% of the 2.5 MPa set point, upheld superheat within ± 0.2 K of the nominal 10 K, and constrained net efficiency operation to a tight 0.20–0.30 band. This reduced operational volatility delivered approximately a 6 percentage-point increase in the annual mean efficiency. Seasonal assessments confirmed robustness across different irradiance regimes, smoother actuator trajectories, minimized overshoot, and disciplined, repeatable TES cycles that avoided thermodynamic penalties associated with overcharge and depletion events. To refine operating performance, a multi-objective GA optimization applied exclusively to DRL-controlled operational data. The resulting pressure-efficiency-temperature Pareto front revealed well-defined non-dominated solutions clustered around 2.55 MPa and 10.1–10.7 K superheat, aligning with the DRL policy's natural operating locus. Physically consistent dominated clusters represented under-pressurized, over-pressurized, or thermally inefficient regimes, confirming that DRL

control already positioned the plant near its global optima. Importantly, the GA also provided decision-makers with a front-wide trade-off mapping, enabling strategic adjustments if priorities shift between efficiency, component stress mitigation, and TES cycling resilience. The integration of intelligent DRL control and post-hoc multi-objective optimization presents a ready-to-deploy technological pathway for overcoming inherent instability in solar-driven ORCs. Beyond delivering substantial average efficiency gains, the approach enhances operational stability, minimizes mechanical and thermal cycling stresses, and extends component service life. By relying solely on measured or simulated plant data without requiring explicit plant physics models for the controller, the method is broadly transferable and compatible with diverse system architectures, TES technologies, and geographic locations experiencing variable solar input. This positions the DRL integrated with GA framework as a high value Supervisory Control and Data Acquisition upgrade strategy for both retrofits and new-build solar-ORC configurations targeting long-term reliability and maximized return on investment.

7. Future research directions

Future work will focus on expanding the controller's adaptability to incorporate direct online learning for evolving solar and load profiles, integrating weather forecasting modules to improve predictive dispatch, and extending the framework to variable-geometry expanders for finer part-load control. Comparative studies against alternative advanced control strategies (most likely model predictive control and proximal policy optimization) will be conducted to benchmark performance and computational cost. Furthermore, coupling DRL control with real-time multi-objective optimization envisioned to enable continuous movement along the Pareto front in response to operator-selected priorities, providing dynamic trade-off management between efficiency, component life, and TES utilization under transient operating conditions.

CRedit authorship contribution statement

Araz Emami: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Resources, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Ata Chitsaz:** Writing – review & editing, Visualization, Supervision, Resources, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Amirali Nouri:** Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Resources, Methodology, Investigation, Formal analysis, Data curation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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